

2024

Potentiating effects of interim tests and metacognitive judgments on future learning

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Potentiating Effects of Interim Tests and Metacognitive Judgments on Future Learning

by

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A thesis submitted to the Department of Psychology
in partial fulfillment of the requirements for the degree of

Master of Science in Psychological Science

UNIVERSITY OF NORTH FLORIDA

COLLEGE OF ARTS AND SCIENCES

May, 2024

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ACKNOWLEDGEMENTS

I would like to dedicate this document to my family, friends, loved ones, and to the wonderful University of North Florida (UNF) staff and faculty. My advisors, Dr. Sara Davis and Dr. Charles Fitzsimmons, have been sources of endless inspiration and have been gracious enough to impart their wisdom to me, and for that I am forever grateful. To the UNF faculty who I have been fortunate enough to interact with and learn from over the course of my academic journey, thank you; your classes have provided me with the tools required to produce this document and to become a lifelong learner. I would also like to extend my genuine appreciation for Delores Irvin and Katie Haft of the Department of Psychology office for their help answering my questions and providing me with the information and tools necessary to comfortably navigate the program.

Thank you to my family, friends, and loved ones. My parents have been more supportive, loving, and understanding than I could have ever hoped for them to be as I completed this leg of my academic journey, and I cannot thank them enough. Thank you to my friends and brothers, who brought levity to otherwise stressful and daunting endeavors. Finally, thank you to Brenda, for always being in my corner. You always made sure that I felt confident in my ability to surmount my fears and to get through the rough periods of my time in the program, and I genuinely could not have made it here without your love and support.

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Abstract

Interpolated tests benefit retention of both tested material and material learned afterward. One proposed mechanism for the potentiating effect of testing involves the learner gleaned metacognitive information from tests regarding how well the material is being learned, curbing overconfidence and influencing future encoding strategies. The current study investigated whether making judgments of learning (JOLs) may similarly induce metacognitive introspection and benefit learning. An online sample of 464 adults from the United States was randomly assigned to one of three interim task conditions and one of three JOL conditions that determined how they would interact with each of the four text passage sections they read. Participants either made no JOLs, one JOL, or four JOLs, and either restudied, completed a test, or completed a brief filler task after each section. All participants were tested on the final section. Analyses revealed that participants who were tested on previous sections performed better on a test of the final section than those who were not tested. Though there was no significant interaction between interim task and JOL conditions, exploratory analyses found that eliciting JOLs eliminated the potentiating effects of testing under certain conditions. Eliciting JOLs may influence future memory and may not be a neutral event in retrieval practice contexts. Future studies are recommended to determine the boundary conditions of JOL reactivity in testing paradigms.

Keywords: Testing, Forward Testing Effect, Metacognition, Judgments of Learning

Potentiating Effects of Interim Tests and Metacognitive Judgments on Future Learning

Tests are a fundamental part of the education systems of most developed countries (Lau, 2022). With this being the case, there has rightfully been a wealth of discussion concerning the utilization of testing in schools since the formal academic integration of standardized testing in the 20th century. Though testing in education is typically viewed purely as a knowledge or aptitude assessment (Roediger & Karpicke, 2006b), it is becoming increasingly clear that taking a test is not a neutral event. Relative to restudying, periodically testing oneself throughout a study session has been shown to be beneficial for long-term retention of the tested material, a phenomenon termed the backward testing effect (McDermott, 2021). Further, periodic testing has also been shown to benefit learning of future, untested material, a phenomenon commonly referred to as the forward testing effect (Yang et al., 2018). Several mechanisms have been proposed for why this forward effect of testing occurs, one of which positing that tests provide the learner with metacognitive information regarding how well the material is being learned, which then curbs overconfidence and advises future study strategies (Chan et al., 2018). In line with this, research has been conducted to examine whether explicitly asking learners to make metacognitive judgments regarding their confidence in their learning of the material has a similar effect (Lee & Ha, 2019), with recent evidence suggesting that making these metacognitive judgments benefits future learning for untested participants, eliminating the forward testing effect (Davis & Chan, 2023).

Beneficial Effects of Testing

The backward testing effect (BTE; or the retrieval practice effect) refers to the robust phenomenon of retrieval practice improving long-term retention of the retrieved material. With the first study into this phenomenon dating back over a century (see Abbott, 1909), it has since

been demonstrated with a variety of materials, including word lists (Rowland, 2014), word pairs (Jacoby, 1978), and text passages (Roediger & Karpicke, 2006a). Similarly, the BTE has also been found to be robust in a number of populations and environments, with testing being shown to be beneficial in laboratory settings and classrooms (Leeming, 2002), as well as in adults (Bishara & Jacoby, 2008), children (Rohrer et al., 2010), and memory-impairment patients (Camp et al., 2000; Schacter et al., 1985), demonstrating its reliability as a learning aid. The typical paradigm utilized in examining the benefits of testing involves splitting participants into two groups: one that studies some material (e.g., word pairs, passages, etc.) twice and one that studies the material once and takes a test on it. After a delay, both groups then take a test covering the material, with the majority of studies finding that the group that had previously engaged in retrieval of the material performs better on the delayed test than the restudy group (see McDermott, 2021 for a review). The benefits of testing can be enhanced further when feedback is given following the initial test (Roediger & Butler, 2011).

Past studies have also found that testing can potentiate future learning, with participants assigned to the test condition performing better on retention of untested material as well. One of the earliest examples of such a finding comes from Tulving and Watkins (1974). In a classic A-B, A-C word pair learning paradigm, the authors found that testing participants on one list of word pairs impacted their learning of another, as their learning of a second list of word pairs was impaired when the first word pair list was not tested on. These early observations would eventually spawn a host of studies that converge on the notion that testing not only enhances retrieval of previously learned information, but also enhances memory for material learned afterward.

This phenomenon, now frequently referred to as the forward testing effect (FTE; Pastötter & Bäuml, 2014) or test-potentiated new learning, has become an emerging topic of research in cognitive and educational psychology. While the effect was first described in earlier BTE research, Szpunar and colleagues' (2008) experiments examining the FTE established the study design frequently used in its investigation. In this study, participants were randomly assigned to different conditions and tasked with studying a series of word lists. Participants either completed tests between word lists on the list that they had just studied, restudied the same word list by having it presented to them again, or completed a short distractor task between word lists. All participants then took a criterial test on the final word list, as well as a cumulative test covering all of the word lists after a filler task. See Figure 1 for a visualization of this paradigm. The results demonstrated a strong beneficial effect of interpolated testing on retrieval accuracy over the other two conditions on both the criterial and cumulative tests. Intrusion rates, or the rate at which participants incorrectly retrieve words that belong to lists other than the one currently being tested, also decreased in the presence of testing. This reduction in intrusion rates indicates that interpolated testing may serve to protect learners against the negative effect of prolonged study sessions on subsequent learning (Kliegl et al., 2015; Szpunar et al., 2008; Wahlheim, 2015).

Much like the BTE, the FTE has been demonstrated in various populations, including undergraduate students (Nunes & Weinstein, 2012), traumatic brain injury (TBI) patients (Pastötter et al., 2013), and both older and younger children (Dang et al., 2021; c.f. Aslan & Bäuml, 2016). Similarly, the FTE has been found using a number of learning paradigms, including learning word lists (Pastötter et al., 2011), word pairs (Kornell et al., 2009), face-name pairs (Weinstein et al., 2011), and categories (e.g., inductive learning; Lee & Ahn, 2018).

Interpolated tests given during a video lecture have been found to improve learning outcomes, encourage notetaking, and discourage task-irrelevant mind-wandering (Szpunar et al., 2013; but see Jing et al., 2016), as well as improve metacognitive calibration between expected and actual performance (Szpunar et al., 2014). Pertinent both to the present study and to the practical implications of FTE research, interpolated testing has been shown to be effective for potentiating learning of educational materials like text passages as well (Healy et al., 2017; Wissman et al., 2011; Zhou et al., 2015).

Test Potentiated Impairment

Interpolated testing, though found to be largely beneficial for later learning, has also been found to be neutral or even detrimental for learning under certain conditions. When learners switch between encoding and retrieval during study, new learning may be impaired relative to simply restudying prior to learning the new material (Karaca et al., 2020). Finn and Roediger (2013) demonstrated this using a face-name-profession learning paradigm in which participants are tasked with learning face-name pairs, either restudying these pairs or being tested on them, then learning professions attached to these face-name pairs. Participants who were tested on the face-name pairs exhibited worse learning of the professions relative to those who restudied the pairs. The borrowed time hypothesis, or the idea that new learning may be impaired when it is intermixed with retrieval due to learners opting to spend study time relearning the tested material rather than learning the new material (Davis & Chan, 2015), was proposed as an explanation for these findings. In a series of experiments, Davis and Chan (2015) found support for the notion that learners may use the metacognitive information gathered through testing to focus their encoding effort, which may be incorrectly focused on relearning the previously studied information over learning the new information when both are presented in a time-constrained

study session. Davis and colleagues (2017) expanded upon this with their findings suggesting that frequently switching between encoding and retrieval during study (as is the case with frequent interpolated testing) impaired new learning, with this impairment increasing with the frequency of switching and being most profound for material studied immediately after retrieval. Seeing as these results were only found when switching between studying and testing and not studying and restudying, these findings suggest that retrieval and encoding are distinct modes of processing and swapping between these modes during study interferes with encoding efficacy (Evans et al., 2015; Tulving, 1983).

Proposed FTE Mechanisms

Despite the seeming robustness and ubiquity of the FTE in different materials, the proposed mechanisms underlying the FTE remain varied, with no single unifying theory and several mutually non-exclusive explanations being put forth (Chan et al., 2018; Yang et al., 2018). For example, Szpunar and colleagues (2008) proposed that interpolated tests benefit future learning through releasing the individual from proactive interference, the detrimental effects of past learning on future learning that naturally build up over the course of an extended study session. This release-from-proactive-interference account can be incorporated into a larger framework which posits that interpolated tests enhance learning outcomes through alteration of the internal contexts that are encoded alongside target information. Proposed mechanisms within this framework, aptly referred to as context theories, suggest that attempted retrieval of information isolates learning events from one another via altering the internal context attached to that information (Pastötter et al., 2011), thus reducing proactive interference from prior study episodes and enhancing memory test performance by allowing the individual to constrain their

search during retrieval to material that has yet to be tested, or material that has no retrieval context associated with it (Thomas & McDaniel, 2013).

A second theoretical framework posits that interpolated testing serves to increase the cognitive resources available for future encoding and was termed by Chan and colleagues (2018) as the resource account or resource theories. This framework, also referred to as the reset-of-encoding account, at least partially attributes the FTE to interim tests increasing available attentional resources through reducing proactive interference from prior learning events and reducing mind wandering during future encoding (e.g., Szpunar et al., 2013). Past research has found that, as a prolonged study session goes on, individuals begin to experience more frequent lapses in their attention that impair learning via a reduction in attentional resources for encoding (Smallwood & Schooler, 2015), with these resources being restored when the individual attempts retrieval (Pastötter et al., 2011). Eliciting interpolated metacognitive judgments may provide a similar benefit, as taking a break from encoding and engaging in the forecasting processes involved may reduce mind wandering and restore the attentional resources that are depleted in the absence of interpolated tasks. However, some resource theories also posit that interpolated tests reduce the likelihood of intrusions from prior material during future retrieval, possibly through providing learners with a degree of closure regarding their mastery of the material that subsequently allows them to free up the cognitive resources that were being used to hold that material in working memory (Chan et al., 2018). It is currently unclear whether providing predictive confidence judgments would have a similar effect, as future-oriented judgments (e.g., asking participants to provide judgments predicting how well they believe they would do on a future test) may cause learners to anticipate a test and prevent them from experiencing closure.

The third explanatory framework centers on the proposition that interim testing causes learning that occurs after the test to be better integrated with either its retrieval cue or the material learned prior to the test (Chan et al., 2018). The latter may occur during the encoding of the new material if the new material is in some way associated with the original pretest material, such as if they are semantically related or share a cue. The new and old material are integrated and facilitate future memory performance by improving the efficacy of cues and the individual's conceptual organization (Carpenter, 2011; Jing et al., 2016). In other words, taking an interpolated test is likely to cause the individual to covertly retrieve the tested material during new learning, facilitating a connection between the old and new material and forming an integrated representation of the material. Similarly, if the new material and its cue are associated with one another, as is the case with a related word pair, interim tests in which individuals are presented with the cue and tasked with guessing the target may end up activating the target during search, priming it for future learning (Kornell et al., 2009). Use of this paradigm with material that is not semantically related (e.g., traditional questions about text material) may still enhance learning through improving retention of the cue itself and promoting future learning of its target if questions are asked prior to study. Attempting to answer questions induces an elaborative memory search that activates concepts related to the answer being searched for, enhancing retention of the question itself as well as any potential answers retrieved during search. For subsequent learning and tests, the elaborative memory search conducted previously serves to improve learning through deeper activation of the semantic network that surrounds that cue than would be afforded by restudying (Kornell, 2014). In the context of the current study, asking participants to make metacognitive judgments regarding the material being learned may facilitate integration of old and new material to the extent that the judgments allow for covert

retrieval to occur, as would be the case if the judgment-eliciting cue includes partial information from the material (Kubik et al., 2022).

Most relevant to the current study is the metacognitive family of theories. There is a large degree of variability in the exact mechanisms posited within this account; however, they converge on the idea that interim testing serves to enhance future learning outcomes by inducing metacognitive introspection within the individual regarding their learning of the material that then enhances future learning (Chan et al., 2018). Testing can potentiate future learning via many avenues under this framework, including providing information about what aspects of the study material will appear on tests to inform future encoding (Wissman et al., 2011), improving learners' awareness of their ability or lack thereof to learn the material presented (e.g., curbing overconfidence; Cho et al., 2017; Szpunar et al., 2014), altering test expectations (Szpunar et al., 2007), or providing information about descriptive aspects of the material (e.g., cue-target relatedness). While the metacognitive family of explanations for the FTE typically involve testing potentiating future learning via changes to the learner's encoding strategy, recent evidence suggests that participants report no qualitative difference in encoding strategies between testing and restudy conditions (Ahn & Chan, 2023). Learners may instead use the metacognitive information gleaned from interpolated tests to inform better *retrieval* strategies (Chan et al., 2020) or may benefit from quantitative changes to their encoding strategy that are not readily apparent to the participant and thus would not appear on a qualitative strategy report, such as allocation of study time (Davis & Chan, 2023; Yang et al., 2017). Similarly, if participants are asked to make predictions regarding how well the material is being learned between study sessions, this may induce metacognitive introspection in much the same way posited here, possibly leading to a change in, or sustaining of, future strategy.

Judgments of Learning

The introspection during study that characterizes the metacognitive account for the FTE may be examined using judgments of learning. Judgments of learning (JOLs) are a learner's predictions of their own performance on some future assessment of memory (Rhodes, 2015). In the monitoring and control framework produced by Nelson and Narens (1990), JOLs are reflective of one type of monitoring process (i.e., one's cognitive assessment of how well the material is being learned) and the information gained from these monitoring processes are used in control processes that are reflected in future study decisions. For example, if a student is studying for an upcoming exam and reflects on how well they have learned the topics they have covered thus far in their study session, the information gleaned from that reflection would let the student know what topics they have struggled to learn and likely need to restudy. Accordingly, learners who are overconfident in their future retention of the material may choose to terminate their study session prematurely, leading to worse performance (Dunlosky & Rawson, 2012). JOLs are elicited either directly after encoding or after some delay. Of special interest in the literature are measures of absolute accuracy and relative accuracy. Absolute accuracy (or calibration) refers to the extent to which one's average degree of confidence as reported by JOLs correlates with their performance on a subsequent memory test; if the average JOL, as measured by a percentage, is higher than the percentage of items correctly recalled on the memory test, then the learner was overconfident in their performance. Relative accuracy (or resolution) is measured on an item-by-item basis and refers to how accurately one's JOLs predict their performance for each item, or how well one's JOLs distinguish between items that are correctly recalled and those which are not (Rhodes, 2015).

Relatively few studies have examined how interim testing influences learners' confidence judgments. In one such study, Szpunar and colleagues (2014) had participants view four sections of a video-recorded lecture. Participants were randomly assigned to one of three conditions: one in which participants received a test after each section of the lecture, one in which participants received a test only after the final section, and one in which participants received no tests. After viewing the final lecture section, all participants made a prediction, using a 0-100% scale, regarding how well they believed they would do on a cumulative test of the lecture content. All participants then took a self-paced cumulative test. Interestingly, and in contrast with what is postulated by the metacognitive account, there was no significant difference between predicted performance in the four interim tests group compared to the group that received no interim tests. The interim test group did, however, show improved actual performance, which, due to the general overconfidence exhibited by untested participants, translated into greater calibration between predicted and actual performance. Across several experiments and using a variety of materials, Yang and colleagues (2017) had participants encode material while either taking tests between lists or not. Participants also had their encoding time measured and provided aggregate JOLs following each list, indicating on a sliding scale how many items they believe they would accurately recall if tested. Untested participants showed decreases in both their confidence judgments and time spent encoding across the lists, while the confidence judgments and encoding time for participants who were exposed to interim tests remained stable. Unlike Szpunar et al. (2014), both groups showed accurate metacognitive calibration between predicted and actual performance.

Judgment of Learning Reactivity

Past literature suggests that eliciting JOLs may induce reactivity in learners and impact future memory performance. Reactivity refers generally to the finding that psychometric measurement is not a neutral event, with participants tending to exhibit changes in their behavior as a result of realizing they are being observed (French & Sutton, 2010). JOL reactivity refers more specifically to the finding that making JOLs is not a neutral memory monitoring task as may be presupposed (Nelson & Narens, 1990), as the act of making a JOL may influence subsequent memory outcomes (Rhodes, 2015). When individuals monitor their own learning, they make judgments regarding how well the information is being learned, then use these judgments in control decisions regarding future study (Ariel et al., 2021; Metcalfe & Finn, 2008). Previous research has found that the effect of these judgments on learning is dependent on both the material being studied and the type of test presented. Several studies have demonstrated a beneficial effect of JOLs on memory performance for related items on a cued-recall criterial test, but no such effect for unrelated items on a cued-recall test or either class of items on a free recall test, suggesting that making JOLs direct learners' attention to aspects of the material (i.e., relatedness) that would otherwise be weakly processed or not processed at all, strengthening the relationship between the two words in the judged pair (Halamish & Undorf, 2022; Myers et al., 2020; Rivers et al., 2021; Soderstrom et al., 2015). Separate from this cue-strengthening view is the cognizant confidence hypothesis, which posits that JOLs serve to draw the learner's attention to their confidence and goals regarding the material, benefiting retention if their confidence is high and their goals conducive to learning (Double et al., 2018). The heterogenous effect of JOL reactivity depending on material type is partially explained via this hypothesis, as participants are generally more confident in their JOLs when studying related word pairs and highly confident

participants are more likely to exhibit positive reactivity in other paradigms (Double & Birney, 2017).

The JOL reactivity literature has focused primarily on the memorial effects of JOLs on retention of the material being judged. That is, the effects of JOLs on memory have typically been examined using cumulative tests that include only the material that has been evaluated by participants using JOLs. However, the forward effect of JOL reactivity (i.e., the possible effect of making JOLs on retention of material learned afterward) has received considerably less attention. As touched on in previous sections, JOLs may engage some of the same mechanisms proposed in the different FTE explanatory frameworks to be engaged by retrieval practice. In one study that investigated the forward effect of JOL reactivity, Lee and Ha (2019) examined the effect of interpolated JOLs on inductive learning outcomes and found that making category-level JOLs and global confidence judgments - but not item-level JOLs - enhanced future learning of painters' painting styles. The authors put forth two possible explanations for their findings. First, the elicitation of category-level and global JOLs and the subsequent retrieval or description failures experienced by participants may have informed participants about the difficulty of memorizing and describing complex information (e.g., painting styles), inducing changes in effort or strategy during future study. Second, the category-level and global JOLs may have provided participants with information regarding learning goals, providing information about the task and what would be expected of them on any subsequent test(s). A separate explanation proposed for why making JOLs may enhance future learning is the covert retrieval hypothesis. This hypothesis posits that the act of making JOLs induces covert retrieval of the learned information, causing participants to unconsciously retrieve the judged material and subsequently producing a similar beneficial effect to the overt retrieval elicited by interpolated testing. For

example, Kubik and colleagues (2022) found that making JOLs using word stems, which would likely result in covert retrieval attempts, benefited future learning relative to restudying; however, the same beneficial effect was not found for JOLs made using complete words, which are less likely to induce covert retrieval. Seeing as JOLs are commonly included in memory paradigms to examine learners' confidence in the material, the presented evidence for JOLs' capacity to impact future memory test performance and past studies' seeming unawareness of this effect call this practice into question. Further, the similarities between interim testing and interim JOLs in their memorial benefits and proposed mechanisms provide an opportunity to investigate whether the processes behind these two effects are identical.

Current Study

The current study sought primarily to assess the metacognitive approach to the FTE by comparing interim testing with untested JOLs that may induce reactivity and subsequent changes in learning strategies. This study builds upon the secondary investigation of JOL reactivity in an FTE paradigm conducted by Davis and Chan (2023). Using text passages, Davis and Chan (2023) conducted experiments to examine whether varying the difficulty of interpolated tests in an FTE paradigm would produce varying effects on criterial test performance. The authors posited that participants taking a more difficult interim test (i.e., a cued recall test) would be less likely to be overconfident in their learning than participants who took an easier test (i.e., a multiple-choice test) or no test, due to more difficult tests being more likely to induce optimization of study strategies for future passages and exertion of more effort during encoding. The authors found support for this hypothesis, with participants in the more difficult test condition exhibiting a larger FTE than participants in the easier test condition. The authors also found evidence of a strategy difference in participants' reading times, as tested participants spent

more time reading the text passages than untested participants. The authors reconducted these experiments, but had all participants provide four aggregate JOLs predicting future test performance following encoding of each passage to explicitly examine whether confidence differed between conditions. Critically, in both experiments in which JOLs were elicited, untested participants and tested participants did not significantly differ in criterial test performance or reading time, eliminating the FTE. The authors posited that asking participants for several JOLs that required specific predictions regarding future performance on tests of different types at different retention intervals induced deep introspection that produced a beneficial reactivity effect in untested participants, inducing strategy change in a process akin to that hypothesized to occur during interim testing. To further investigate this finding, the current study employed three JOL conditions (no JOLs, one JOL, four JOLs) attempted to replicate the result found by Davis and Chan (2023) and to examine whether providing a single JOL that would seemingly require less metacognitive reflection would still produce the observed FTE elimination effect. Additionally, the current study sought to determine whether making JOLs may provide potentiating effect to restudy opportunities, such that learners who make JOLs predicting their future recall can then check how well they remembered the material when given the opportunity to restudy the material, inducing strategy change or maintenance and benefiting future recall.

As stated, retrieval practice provides learners with an opportunity to assess how well their learning strategies are working, informing how they approach future study. It has been proposed that JOLs may reactively produce a similar effect, prompting metacognitive introspection that provides individuals with information about their learning. The current study proposes that the extent to which this metacognitive introspection is induced may inform the reliability or quality

of strategy assessments and the control decisions made from those assessments. Therefore, the current study hypothesized that the FTE would be largely eliminated when comparing tested participants and untested participants that made several JOLs, but that the FTE would persist when comparing tested participants with untested participants that only made a single JOL.

Metacognitive explanations for the FTE also suppose that the metacognitive information gleaned from testing serves to curb the overconfidence that is often exhibited by learners (Chan et al., 2018; Karpicke, 2009). JOLs provide information regarding how well participants believe they will do on future tests of memory. While these judgments are typically used in conjunction with memorial tests to determine learners' forecasting accuracy (Rhodes, 2015) and may be beneficial in examining the confidence curbing account for the FTE (Chan et al., 2018), the potential for JOLs themselves to influence memory performance questions their viability in testing paradigms. Given the results of Davis and Chan (2023), it may not be appropriate to elicit JOLs in FTE paradigms to examine whether confidence is being affected by interim testing if the effect of testing is eliminated by their inclusion. Thus, the current design should also provide further clarification regarding the elicitation of JOLs and whether it is appropriate to include them in FTE paradigms going forward.

Method

Participants

For the purposes of the present study, 360 participants were planned to be recruited via Prolific (www.prolific.co). Prolific is an online participant curation platform designed to generate higher quality data through participant screening, while also providing participants with fair financial compensation. While similar in concept to Amazon's Mechanical Turk platform, Prolific is geared solely toward conducting research and thus comes with several features that

improve usability for both researchers and participants, including greater security, ease of use for complex designs, sampling options, and more. Only participants from the United States were recruited in the study. Additionally, as a way of ensuring participant quality, participants were only recruited if they had previously participated in at least 20 other studies through Prolific. The results of a meta-analysis conducted by Chan et al. (2018) revealed an overall effect size of Hedge's $g = .75$ when comparing criterial test performance between tested and untested participants across the test-potentiated new learning literature that was selected for analysis. A pilot study using the materials utilized in the current study was preregistered through Open Science Framework (OSF; <https://osf.io/vahsg>) and conducted prior to data collection. Preliminary pairwise comparisons of criterial test performance between participants who were tested between reading text passage sections and those that were not tested in this pilot study revealed an effect size of $d = 0.52$ (see Appendix A). An estimated effect size of $d = 0.65$ was determined with consideration given to both the results of the pilot study and the results of the meta-analysis conducted by Chan et al. (2018). To detect a mean difference at an effect size of $d = 0.65$ with 80% power using an alpha of .05, a priori power analyses conducted using G*Power suggest a sample size of 39 participants per between-subjects condition. Given the current study's 3 x 3 between-subjects factorial design and rounding to 40 participants per group, a total sample size of 360 participants who completed both parts of the study was deemed appropriate for the current study.

Current Sample

Data were collected from a total of 495 participants via Prolific over the course of two waves of data collection. Of these participants, 196 did not have observations for criterial section JOLs due to experimenter error with the first wave of data collection. Seeing as this was the only

notable difference between the two waves, data from the two participant batches were combined for the analyses reported here. Data from 31 participants were excluded from further analyses. Four participants were excluded for not following instructions (i.e., providing inappropriate responses to test questions such as random words or phrases), five for refusing to interact with the task (i.e., skipping through test questions, as evidenced by null responses and inordinately brief question display times), one for participant error, four for admitting to cheating on either the criterial or cumulative test, eight for responding to every presented JOL prompt with either 0% or 100%, and nine for responses that appeared to indicate use of generative AI or a search engine on either part of the survey (see Figure 2 for an example). After these exclusions, 464 participants (232 male, 223 female, 6 nonbinary, 1 agender, 2 undisclosed) were included in the final sample. The final sample had a mean age of 41.86 years ($SD = 13.01$) and, of these participants, 6.5% of the sample identified as Asian, 11.9% as Black or African American, 2.4% as Hispanic or Latino, 0.4% as Native Hawaiian or Pacific Islander, 72.6% as White, 5.1% as multiracial, and 1.1% of participants chose to keep their racial/ethnic identity undisclosed or identified as an unlisted race/ethnicity. Participants received \$2.00 for completing the first part of the study and \$1.50 for completing the second, for a total of \$3.50 in compensation for those that completed both parts. Part 2 was intended to be completed approximately 24 hours after completion of Part 1. As such, the study instructions in Prolific included asking participants to ensure that they would be available approximately 20-28 hours after their participation in Part 1. In a follow-up message sent to all participants that completed Part 1, participants were asked to complete Part 2 as close to 24 hours after their completion of Part 1 as reasonably possible. The average time between completing Parts 1 and 2 was 23.59 hours ($SD = 4.35$; Table 1).

Design

A 3 (Interim Task: No Test, Restudy, Test) x 3 (JOLs: No JOLs, Single, Multiple) between-subjects factorial experimental design was utilized in the current study. The interim task that participants completed and the JOLs elicited were manipulated, with participants being randomly assigned to one of the three interim task conditions and one of three JOL conditions. The primary dependent variable measured in the present study was criterial test performance, which was measured by the proportion of correct responses on the final section test. For the purposes of the second research question, JOLs were evaluated as well. The current study and primary analyses were preregistered through OSF (<https://osf.io/z3qkj>).

Materials

The following materials and procedure were presented using Qualtrics surveys that were made available to participants via an anonymous link upon choosing to participate on Prolific. Prior to data collection from the full batch of participants, both Qualtrics surveys were made available to a test batch of 20 participants via Prolific to ensure the survey flow worked properly and participants were correctly being randomly assigned to their interim task and JOL conditions. One test batch of 20 participants was run at each wave of data collection. Preliminary examinations of the data showed that random assignment procedures worked as intended and pages were being appropriately displayed to participants in both test batches. As such, the data from these test batches were included in the reported analyses.

Text Passages and Questions

The text passages used in the present study were adapted from passages that appear in the introductory OpenStax Psychology textbook (Spielman et al., 2014) and were originally utilized by Daley and Rawson (2019). Each passage pertained to a different topic related to memory,

with passages covering the topics of short-term memory (109 words; Flesch-Kincaid Grade Level: 12.7), long-term memory (102 words; Flesch-Kincaid Grade Level: 9.5), explicit memory (104 words; Flesch-Kincaid Grade Level: 13.1), and retrieval (101 words; Flesch-Kincaid Grade Level: 11.3). Passages were minorly edited for clarity and to ensure that each passage was of comparable length. The text passages were presented in the same order for each participant. Four short-answer questions were prepared for each of the four text passages (Appendix B).

Participants responded to test questions by typing their answer into a textbox shown under the question. Many of the test questions used in the present study were adapted from the cued recall questions used by Daley and Rawson (2019), with questions being constructed to ensure that they could be adequately answered using 1-3 words (e.g., Q: “What type of memory stores information about events we have personally experienced?” A: episodic memory). Test questions were presented one at a time and were self-paced, as participants were given an unlimited amount of time to answer them. Responses to each test question were typically coded as either correct or incorrect when calculating performance; however, one question in the criterial section afforded participants the opportunity to receive partial credit, as it asked participants to list three key terms learned in the passage and they received the equivalent of one third of a correct response for each term correctly listed. This was the only question across the four tests where partial credit was appropriate.

These materials are novel and created for the purpose of this study. As such, a pilot study was conducted using the passages and test questions to determine whether they would be capable of exhibiting the FTE. The research questions of the current study are only exploratory given the presence of a traditional FTE, as it would not be appropriate to explore the mechanisms behind the FTE and whether JOLs may eliminate it if the materials used are incapable of producing it in

a traditional testing paradigm. A brief inspection of the pilot results showed the passages and questions to be suitable for examining the proposed research questions. See Appendix A for the results of this pilot study.

Judgments of Learning (JOLs)

Participants in the present study were exposed either to no JOLs, one JOL, or four JOLs after reading each section and completing a brief filler task. The Single JOL condition asked participants to make a prediction regarding how well they feel they could perform on a test regarding the previously read passage, presenting participants with the prompt: “From 0% correct to 100% correct, how well do you believe you would do on a test concerning the passage you just read? Please answer using the sliding scale below.” The Multiple JOL condition were adapted from Davis and Chan (2023) and asked for four predictions in relation to specific types of tests and retrieval timeframes (e.g., “From 0% correct to 100% correct, how well do you believe you would do on a multiple choice test concerning the passage you just read if you were given one 25 minutes from now? Please answer using the sliding scale below.”; Appendix C). Participants responded to the JOL prompt(s) by interacting with a sliding scale from 0-100, with 0 indicating participants’ beliefs that they would perform very poorly and 100 indicating participants’ beliefs that they would perform very well.

Procedure

The present study was granted approval from the Institutional Review Board (IRB#: 1997156-1) at the University of North Florida and was conducted in line with the American Psychological Association’s Ethical Principles and Code of Conduct (American Psychological Association, 2017). Participants were given an opportunity to sign up for the study via Prolific and the platform provided them with a link to the Qualtrics survey utilized in the study. No

directly identifiable information was collected by the survey and responses were anonymized. Prolific IDs were collected to ensure that participants who had difficulty using the code presented at the end of the survey received compensation.

Upon agreeing to the informed consent form, participants were shown a paragraph of instructions informing them that they would be presented with several text passages. The instructional paragraph informed participants that, after reading each text section, they would either take a test on the passage, reread the passage, or complete another task, decided at random. Additionally, participants were implored to not take notes when reading the text passages, nor use any search engines when answering test questions. Participants were assigned to one of three between-subjects JOL conditions, determining whether they provided no JOLs (No JOLs Condition), one JOL (Single Condition), or four JOLs (Multiple Condition) after reading each text passage. Participants were also assigned to one of three between-subjects interim task conditions. Upon completion of their JOL condition task for each passage, participants either completed another 30 seconds of a filler task (No Test Condition), were given an opportunity to restudy the passage for 30 seconds (Restudy Condition), or took a four-question short answer test on the passage that they just read (Test Condition).

During the initial study phase, participants were shown one text passage at a time. The instructions informed them that they could spend as much time as they would like to read the passage during this study phase, though the button to advance to the next page of the survey did not appear until at least five seconds had passed. After reading each passage, all participants completed a brief word generation filler task in which they were shown a lengthy English word (e.g., RADIOACTIVE) and asked to generate as many words using the letters found in the long word as they could in 30 seconds, typing the generated words into a textbox. Participants then

completed the tasks relevant to their assigned condition. See Figure 3 for a visualization of this procedure. As an example, upon completing the short filler, participants in the Test + Single JOL condition responded to the single JOL before they completed the test that covered the material in the passage that they had just read. Regardless of condition, all participants completed a short-answer test for the final passage - the criterial test. Performance on the criterial test serves as the *criterion* by which test-potentiated new learning is measured. Upon completion of the criterial test, participants were asked several demographic questions pertaining to their age, gender, race, education, and whether they had prior experience with either psychology or the specific material presented in the text passages.

After approximately 24 hours, all participants who completed this first part of the study were invited to participate in the second part in which they were presented with a cumulative test that covered the material in all four passages. Participants were informed of this second part prior to completing the first part. Participants who completed Part 1 were also sent a message via Prolific thanking them for their participation and reminding them to complete the second part of the study as near to 24 hours after their completion of Part 1 as possible. Participants were not informed that the second part of the study would consist of a cumulative test prior to starting it. Part 2 began with participants completing another informed consent form and reading an additional instructional paragraph. The instructional paragraph informed them that they would be taking a test covering the material they read in Part 1 and requested that they do not use any search engines when answering the questions. The cumulative test presented participants with the same 16 questions from the four interim tests one at a time in random order. After participants completed the cumulative test, they were debriefed and presented with a code that they copied and pasted into Prolific to receive compensation for participating in the study.

Results

Unless otherwise specified, all data analyses reported here were conducted using the JASP v0.18.3 open-source statistical analysis program for Windows operating systems (JASP Team, 2024). As stated previously, the primary analyses reported here were preregistered along with the study design through OSF (<https://osf.io/z3qkj>).

Model Assumptions

The primary research questions of the current study were explored using between-subjects analyses of variance (ANOVAs). The data must meet a number of assumptions to be appropriate for use with ANOVA analyses. The independent variable(s) must be categorical and consist of at least two levels or conditions, the dependent variable must be quantitative in nature, the dependent variable must be normally distributed, and there should be homoscedasticity for the dependent variable between groups. The first assumption was met by the study design, as participants were randomly assigned to one of three between-subjects interim task conditions and one of three between-subjects JOL conditions; both independent variables were categorical and consisted of three levels each. The primary dependent variable, the number of correct responses to the criterial test, is quantitative in nature and ranged from 0 to 4 for each participant. The accuracy variable used in these analyses, calculated as a simple proportion of correct responses, meets this assumption as well.

To examine whether the quantitative variables of interest were normally distributed, Q-Q plots were constructed and examined (Ford, 2015). Examining the Q-Q plots for most of the quantitative variables of interest (reading times, criterial/interim test performance, JOLs, and cumulative test performance) revealed these variables to be approximately normally distributed; however, it is worth noting that criterial and interim test performance are both subject to ceiling

and floor effects due to the limited variability of these variables caused by only four questions being used for each section test. Similarly, intrusion rates were shown to not be normally distributed in the data given the limited variability. This being said, ANOVAs are robust against moderate departures from normality (Blanca et al., 2017).

Finally, homoscedasticity of the dependent variables of interest between conditions was analyzed using Levene's Test for Equality of Variances. Levene's Test is conducted with each ANOVA, and if the p-value generated by the test is not below .05, then it can be reasonably assumed that the variances between groups do not significantly differ, and the assumption of homoscedasticity is met. The results of this test were reported where the results of any ANOVAs conducted are presented below. In instances in which the homogeneity of variances assumption was violated, a Welch correction was applied to the ANOVA and those results are reported. The Welch correction adjusts the degrees of freedom of the ANOVA and is recommended in most situations in which the homogeneity of variances assumption is violated (Glantz et al., 2016). Additionally, in cases in which the Welch-corrected analysis reveals a significant result, pairwise comparisons were performed using Games-Howell post hoc tests. Games-Howell post hoc tests do not assume equal variances and are thus recommended over Tukey's HSD test in situations in which the homogeneity of variances assumption is violated (Lee & Lee, 2018).

Interim Test Performance

Testing for differences between JOL conditions within the Test condition on Section 1 interim test performance allowed for examination of whether there were any baseline differences in performance or learning ability prior to the manipulation having an effect in later sections. Levene's Test for Equality of Variances revealed no violations of the homogeneity of variance assumption, $F(2, 151) = 0.29, p = .75$. Contrary to expectations, a one-way ANOVA revealed

that participants significantly differed in Section 1 test performance by JOL condition ($F(2, 151) = 5.50, p = .005, \eta_p^2 = .07$), with post hoc comparisons using Tukey's HSD test revealing that participants who made no JOLs ($M = 0.75, SD = 0.24$) performed significantly better than participants who made multiple JOLs ($M = 0.59, SD = 0.25; p = .004, d = 0.65$). Participants who made one JOL ($M = 0.66, SD = 0.25$) did not significantly differ from participants who made multiple JOLs ($p = .44$) nor from participants who made no JOLs ($p = .094$). Follow-up exploratory analyses were conducted to determine whether these conditions varied by prior experience with the material. There was no significant relation between tested participants' JOL condition and whether they had previously taken a psychology course ($\chi^2(2, N = 152) = 0.99, p = .61$) nor was there an association between JOL condition and whether they had prior experience with the topics presented in the text passage ($\chi^2(2, N = 154) = 1.50, p = .47$).

A one-way ANOVA examining differences in Section 2 interim test performance between JOL conditions was conducted. Homogeneity of variances was not violated in this analysis ($F(2, 151) = 2.48, p = .087$). There was no significant effect of JOL condition on interim test performance for Section 2 ($F(2, 151) = 2.68, p = .072, \eta_p^2 = .03$). A one-way ANOVA investigating differences by JOL condition for Section 3 interim test performance was conducted as well. The assumption of homogeneity of variance was not violated ($F(2, 151) = 2.31, p = .10$). The results showed no significant differences in Section 3 test performance by JOL condition ($F(2, 151) = 0.78, p = .46, \eta_p^2 = .01$).

Criterial Test Performance

A 3 x 3 between-subjects factorial ANOVA was conducted to examine how criterial test performance varied between conditions and to determine whether eliciting JOLs eliminates the FTE in a factorial design. Levene's Test showed no significant differences between conditions,

$F(8, 455) = 1.67, p = .10$, meeting the assumption of homoscedasticity. The 3 (between-subjects Interim Task condition: No Test, Restudy, Test) x 3 (between-subjects JOL condition: No JOLs, Single, Multiple) between-subjects factorial ANOVA showed a significant main effect of interim task condition on criterial test accuracy, $F(2, 455) = 8.58, p < .001, \eta_p^2 = .04$. As can be seen in Figure 4, post hoc comparisons conducted using Tukey's Honest Significant Difference (HSD) test revealed that participants in the Test condition ($M = 0.70, SD = 0.30$) showed significantly higher criterial test accuracy than participants in the No Test ($M = 0.55, SD = 0.33; p < .001, d = 0.44$) and Restudy ($M = 0.57, SD = 0.35; p = .003, d = 0.37$) conditions. There were no significant differences in performance between the No Test and Restudy conditions ($p = .37$), indicating that, across JOL conditions, having a chance to restudy the first three study sections did not benefit future learning relative to completing a filler task. Additionally, there was no significant main effect of JOL condition, $F(2, 455) = 0.11, p = .89, \eta_p^2 < .01$. Across interim task conditions, making JOLs for each section did not significantly impact future memory performance. Finally, contrary to the primary hypothesis, there was no significant interaction between interim task and JOL conditions in the present analyses $F(4, 455) = 0.56, p = .70, \eta_p^2 = .01$. Though an FTE was found overall, the lack of an interaction between participants' interim task and JOL conditions runs contrary to the primary hypothesis that there would be no difference in performance between tested and untested participants who made multiple JOLs.

Given that the factorial analysis did not show the expected interaction effect, exploratory analyses were conducted to determine whether an FTE was present within each of the JOL conditions. Davis and Chan (2023) found that untested and tested participants that both made multiple JOLs did not differ in criterial test performance, despite finding an FTE with the same materials when participants did not make JOLs. Seeking to replicate these results and investigate

the a priori hypotheses of the study, we conducted one-way between-subjects ANOVA analyses examining criterial section performance by interim task condition within each of the JOL conditions. Starting with the No JOLs condition, Levene's Test showed homogeneity of variance to be violated when comparing performance between interim task conditions, $F(2, 164) = 4.40, p = .014$. The ANOVA revealed a significant effect of interim task condition on criterial test accuracy, $F(2.00, 107.93) = 5.43, p = .006, \eta_p^2 = .06$. Post hoc comparisons using Games-Howell tests revealed that, among participants who made no JOLs, tested participants ($M = 0.71, SD = 0.28$) performed significantly better than untested participants ($M = 0.56, SD = 0.34; p = .048, d = 0.45$) and participants who restudied each section ($M = 0.53, SD = 0.36; p = .011, d = 0.55$). These results demonstrate a traditional FTE; participants' future learning outcomes benefited from interpolated testing during study, such that participants who were tested on each section prior to the criterial test performed better than those who were not and those who restudied each section.

A one-way ANOVA examining performance within the Single JOL condition was conducted. The homogeneity of variances assumption was not violated, $F(2, 149) = 1.13, p = .33$. The analysis revealed there to be significant differences in criterial test performance by interim task condition when a single JOL was made, $F(2, 149) = 3.96, p = .021, \eta_p^2 = .05$. Post hoc comparisons using Tukey's HSD test revealed that tested participants ($M = 0.72, SD = 0.32$) performed significantly better than untested participants ($M = 0.53, SD = 0.34; p = .017, d = 0.55$), while neither condition significantly differed from participants who restudied the material ($M = 0.60, SD = 0.35; ps > .20$) when making a single JOL.

Finally, another between-subjects one-way ANOVA examining criterial test performance within the condition that made multiple JOLs was conducted. The homogeneity of variance

assumption was not violated, $F(2, 142) = 1.27, p = .28$. Tested participants ($M = 0.67, SD = 0.30$) did not significantly differ in their criterial test performance from No Test ($M = 0.56, SD = 0.33$) or Restudy ($M = 0.59, SD = 0.33$) participants, $F(2, 142) = 1.23, p = .30, \eta_p^2 = .02$. These results replicate the pattern that was found in Davis and Chan (2023; e.g., an FTE was found when examining participants who made no JOLs, but the FTE was eliminated when examining participants who made four JOLs) with the important distinction that tested participants seemed to perform numerically worse when making four JOLs relative to when they make no JOLs rather than untested participants performing better when making multiple JOLs relative to making none.

Intrusions

To examine the possibility that testing benefits future learning through guarding the individual against the detrimental effects of proactive interference (Szpunar et al., 2008), the current analyses examine intrusion rates across the four sections. There are two general categories of intrusions: intralist intrusions and extralist intrusions. Intralist intrusions refer to instances in which participants responded to a test question with an incorrect term or response that was presented within the *same* study section, whereas extralist intrusions refer to instances where the participant's incorrect response was presented in a *prior* study section. Total intrusions were calculated by summing these values. Criterial section intrusion rates were examined using a 3 x 3 between-subjects factorial ANOVA, while interim test intrusions were examined using several one-way ANOVAs to examine possible differences in intrusion rates between JOL conditions. Given that the purpose of these analyses was to specifically examine the possibility that testing protects against proactive interference caused by prior study, the current analyses focus solely on extralist intrusions.

Levene's Test was conducted with each ANOVA reported here to examine the assumption of homoscedasticity. Section 1, being the first section of the passage that participants encountered, was incapable of having extralist intrusions and thus will not be explored further. Section 2 intrusions showed homogeneity of variance between JOL conditions, as evidenced by Levene's Test, $F(2, 151) = 1.17, p = .31$. Section 2 did not show significant differences in intrusion rates by JOL condition, $F(2, 151) = 0.30, p = .74, \eta_p^2 < .01$. Levene's Test produced a significant result when examining Section 3 extralist intrusion rates, indicating heterogeneity of variances between JOL conditions, $F(2, 151) = 7.50, p < .001$. There were no significant differences in Section 3 extralist intrusions by JOL condition, $F(2.00, 88.02) = 1.72, p = .19, \eta_p^2 = .02$.

A 3 (between-subjects Interim Task condition: No Test, Restudy, Test) x 3 (between-subjects JOL condition: No JOLs, Single, Multiple) factorial ANOVA was conducted examining criterial section extralist intrusions and Levene's Test revealed the assumption of homogeneity of variances to be violated ($F(8, 455) = 15.45, p < .001$). The factorial ANOVA showed significant main effects of interim task condition, $F(2, 455) = 11.85, p < .001, \eta_p^2 = .05$. A comparison of average intrusion rates by condition can be seen in Figure 5. There were no main effects of JOL condition ($F(2, 455) = 2.67, p = .070, \eta_p^2 = .01$) nor an interaction effect between interim task and JOL condition ($F(4, 455) = 0.58, p = .68, \eta_p^2 = .01$) on criterial section extralist intrusion rates. Given that the Welch correction cannot be applied to factorial ANOVAs and that the interaction between independent variables was not significant ($F(4, 393) = 1.61, p = .17, \eta_p^2 = .02$), a one-way ANOVA and subsequent post hoc comparisons were conducted to examine the corrected effects of interim task condition on criterial test intrusion rates. The one-way ANOVA examining extralist intrusions by interim task condition was again significant ($F(2.00, 279.14) =$

17.82, $p < .001$, $\eta_p^2 = .05$). Post hoc comparisons using a Games-Howell test revealed that participants in the Test condition ($M = 0.07$, $SD = 0.44$) responded with significantly fewer extralist intrusions on the criterial test than participants in either the No Test ($M = 0.38$, $SD = 0.77$; $p < .001$, $d = 0.44$) or Restudy ($M = 0.45$, $SD = 0.88$; $p < .001$, $d = 0.54$) conditions. The intrusion rates between the Restudy and No Test conditions did not significantly differ ($p = .72$).

Judgments of Learning

As stated, some participants did not have JOL observations prior to the criterial test due to experimenter error in the first wave of data collection. With this in mind, and using the same exclusion criteria described previously, we have criterial section JOL data from 180 participants who were randomly assigned to one of the two JOL conditions that provided JOLs and one of three interim task conditions. For the purposes of the current analyses, the JOL pertaining to how well the participant believed they would perform on a short answer test completed immediately was used for participants who made multiple JOLs per section. It is worth mentioning again that the JOLs provided by participants were their predictions for their future performance on a hypothetical test covering the material they had just read. The prompts used were similar both across the two JOL-providing conditions (predicting future test performance) and within the multiple JOLs condition (being variations of the same prompt varying by timeframe of retrieval and type of test). A 3x2 between-subjects factorial ANOVA was conducted to examine how participant JOLs varied across conditions on the criterial section. A Levene's Test conducted with the ANOVA revealed no significant differences in variance between conditions, $F(5, 174) = 1.37$, $p = .24$, meeting the assumption of homoscedasticity. As observable in Figure 6, the ANOVA revealed no significant main effect of interim task condition ($F(2, 174) = 0.93$, $p = .40$, $\eta_p^2 = .01$), no significant difference between average JOLs made in the Single versus Multiple

JOL conditions ($F(1, 174) = 0.38, p = .54, \eta_p^2 < .01$), and no significant interaction between interim task and JOL condition ($F(2, 174) = 0.16, p = .85, \eta_p^2 < .01$). Contrary to the assumption of the metacognitive framework that testing benefits future learning at least in part due to interim tests curbing learners' confidence, these results suggest that criterial section JOLs did not differ between conditions in the current sample. As such, any differences in learners' predictions of their future performance between conditions was not captured by the current analysis.

Reading Time

First, reading time for the first text passage was analyzed to ensure that there were no baseline differences in reading time between conditions. For the purposes of analyzing study time, outlying reading times that were above the value equivalent to the $IQR * 1.5 + Q3$ were excluded from these analyses for each section. Potential differences in Section 1 reading time were explored using a 3 x 3 factorial ANOVA. Levene's Test showed no significant violation of homogeneity of variance, $F(8, 415) = 1.38, p = .20$. The results of the 3 (between-subjects Interim Task condition: No Test, Restudy, Test) x 3 (between-subjects JOL condition: No JOLs, Single, Multiple) factorial ANOVA revealed no significant main effect of interim task condition ($F(2, 415) = 1.16, p = .32, \eta_p^2 = .01$), no significant main effect of JOL condition ($F(2, 415) = 0.71, p = .49, \eta_p^2 < .01$), and no significant interaction between the interim task and JOL conditions ($F(4, 415) = 1.84, p = .12, \eta_p^2 = .02$). These results indicate that reading time for Section 1 did not differ by condition and that participants' self-paced study times were roughly equal prior to manipulation.

As with prior analyses, criterial section reading time was examined using a 3 x 3 factorial ANOVA. Reading time may be used as a proxy for study effort or a quantitative measure of one's encoding strategy, with higher reading times potentially indicating that the participant is

putting more effort or time into encoding the material. This being said, the interpretation of time spent on the survey page as reading time (and subsequently as a measure of study effort) should be viewed relatively lightly in the context of the current study given that data was collected online. Participants completing online studies are typically not providing the study with their full attention and may be prone to getting distracted, thus making time spent on the page not necessarily an accurate measure of time spent reading/studying the material. Levene's Test showed no significant violations of homoscedasticity, $F(8, 419) = 1.68, p = .10$. The 3 (between-subjects Interim Task condition: No Test, Restudy, Test) $\times$ 3 (between-subjects JOL condition: No JOLs, Single, Multiple) factorial ANOVA revealed a significant main effect of interim task condition on criterial section reading time, $F(2, 419) = 4.52, p = .011, \eta_p^2 = .02$. As seen in Figure 7, post hoc comparisons conducted using Tukey's HSD test revealed that participants in the Test condition ($M = 37.17, SD = 21.75$) spent significantly more time reading the criterial section than participants in the No Test ($M = 29.95, SD = 20.44; p = .015, d = 0.34$) and Restudy ($M = 30.83, SD = 18.16; p = .047, d = 0.28$) conditions, with no significant difference being found between the No Test and Restudy conditions ($p = .90$). There was no significant main effect of JOL condition on criterial section reading time, $F(2, 419) = 0.29, p = .75, \eta_p^2 < .01$. A Pearson's correlation conducted between criterial section reading time ($M = 32.26, SD = 19.97$) and test performance ($M = 0.61, SD = 0.33$) showed reading time and accuracy to be significantly positively correlated in the current sample, $r(426) = .18, p < .001$.

To further assess this possibility, an exploratory mediation analysis examining whether the aforementioned effect of interim task condition on criterial test performance acts through reading time was conducted and is reported in Appendix D. The factorial ANOVA results also showed a significant interaction between the interim task and JOL conditions impacting criterial

section reading time, $F(4, 419) = 4.52, p = .001, \eta_p^2 = .04$. Additional post hoc comparisons conducted using Tukey's HSD test were performed, revealing that, specifically among participants who made no JOLs, Test condition participants ($M = 43.12, SD = 20.44$) spent significantly more time studying the criterial section than participants in the No Test ($M = 26.02, SD = 17.74; p < .001, d = 0.86$) and Restudy ($M = 28.03, SD = 17.53; p = .003, d = 0.76$) conditions. Interestingly, Test condition participants who made no JOLs also spent significantly more time reading than No Test participants who made multiple JOLs ($M = 29.57, SD = 19.81; p = .028, d = 0.68$), Test participants who made multiple JOLs ($M = 29.99, SD = 19.94; p = .050, d = 0.66$), and Restudy participants who provided one JOL ($M = 29.30, SD = 17.65; p = .022, d = 0.70$). No other post hoc comparisons reached significance ($ps > .13$). While the results of this interaction primarily show both that the difference in study time holds true in a traditional FTE paradigm with no JOLs and that the main effect may be driven by study time differences within the condition that made no JOLs, initial interpretations of the final three comparisons are less clear.

Cumulative Test Performance

As noted, the cumulative test consisted of all 16 questions used in the first part of the study presented in a random order. As such, responses were coded similarly, including the one question from the criterial section that afforded the opportunity for partial response credit. In addition, it is worth noting that only participants who took the test between 20-28 hours after their completion of Part 1 were included in these analyses. The desired delay was approximately 24 hours between parts and participants were instructed to complete Part 2 as close to this 24-hour delay as possible. Participants who completed Part 2 of the survey outside of these constraints were still compensated and the survey remained open for an additional 24 hours.

Though not pertinent to the primary questions of the study, the final analyses reported here examined whether a traditional backward testing effect (i.e., taking a test during study benefits future performance on tests covering that same material relative to restudying or doing nothing) was found in the current study, as well as whether making JOLs perhaps had a lasting effect on memory performance.

Levene's Test showed no violations of the homogeneity of variance assumption when examining cumulative test performance between conditions ($F(8, 316) = 0.89, p = .53$). A 3 (between-subjects Interim Task condition: No Test, Restudy, Test) x 3 (between-subjects JOL condition: No JOLs, Single, Multiple) factorial ANOVA revealed a significant main effect of interim task condition on cumulative test performance, $F(2, 316) = 17.64, p < .001, \eta_p^2 = .10$. Post hoc comparisons using Tukey's HSD test found that participants in the Test condition ($M = 0.53, SD = 0.22$) performed significantly better on the cumulative test than participants in the No Test condition ($M = 0.36, SD = 0.20; p < .001, d = 0.81$), as can be seen in Figure 8. Additionally, tested participants performed significantly better than participants who restudied the material during Part 1 ($M = 0.46, SD = 0.21; p = .043, d = 0.33$). Participants who restudied the material also performed significantly better on the cumulative test than participants in the No Test condition ($p = .001, d = 0.48$). Neither the main effect of JOL condition on cumulative test performance ($F(2, 316) = 0.45, p = .64, \eta_p^2 < .01$) nor the interaction between the two independent variables ($F(4, 316) = 0.77, p = .55, \eta_p^2 = .01$) were significant. These results suggest that the elicitation of JOLs did not have lasting impacts on memory of the text passages in the study. However, testing was shown to be beneficial for memory on both the criterial and cumulative tests. Seeing as the current study sought primarily to determine the potential benefits of testing and eliciting JOLs during a study session for *future* learning, the results of a

cumulative test occurring after the study session are not relevant to the questions of the current study and will not be discussed further.

Discussion

The current study sought to extend the findings of Davis and Chan (2023) and determine how eliciting JOLs from learners in a traditional FTE paradigm would impact performance on a criterial test. The primary question which underpinned this study stemmed from the metacognitive framework of explanations for the FTE, which posits that interpolated tests improve future test performance through curbing overconfidence and providing the learner with metacognitive information regarding how well the material is being learned, which is then used to inform future encoding strategies (Chan et al., 2018). Based both upon this framework and the preliminary findings reported by Davis and Chan (2023), the current study investigated whether making JOLs may eliminate the FTE through boosting the performance of untested participants, as the elicitation of JOLs may provide learners with the same chance for metacognitive introspection that is afforded by testing and results in strategy change or maintenance over the course of study. In accordance with this goal, participants in the current study were recruited online and asked to read four sections of a text passage. After each of the first three sections, based on their randomly assigned condition, participants were asked to make one JOL predicting how well they believed they would do on a test of the section, make four JOLs making more specific performance predictions, or make no JOLs. Additionally, participants were asked to either take a brief cued-recall test on the section, restudy the section for 30 seconds, or complete a 30-second filler task. After the final section, all participants completed the brief cued-recall test for that section. Finally, after a 24-hour delay, participants were asked to complete a cumulative test covering all four sections.

Contrary to the primary hypothesis, there was no significant interaction between the interim task and JOL conditions when examining criterial test accuracy. Given that a priori power analyses were conducted with individual pairwise comparisons in mind, it is possible that the current study did not have the power to detect non-crossover interaction effects. As such, exploratory pairwise comparisons were conducted between interim task conditions within each JOL condition. These analyses replicated the findings of Davis and Chan (2023); tested participants outperformed untested participants and those that restudied the material when no JOLs were elicited, exhibiting a traditional FTE, but did not outperform either condition when four JOLs were elicited. Conflicting with this past work, however, is the mechanism by which the FTE was eliminated. In line with the predictions of the metacognitive framework, Davis and Chan (2023) found that eliciting JOLs assumedly caused untested participants to engage in metacognitive introspection similar to that purported to be induced by interim testing, boosting the performance of untested participants to be on par with that of tested participants. Conversely, in the current study, the mean criterial test performance for the tested participants who made four JOLs was numerically lower than the performance of tested participants who made one or no JOLs, while the performance of untested participants who were not given restudy opportunities stayed consistent regardless of how many JOLs were elicited. As such, the FTE elimination effect observed here was seemingly caused by tested participants performing worse when making multiple JOLs, not by untested participants performing better.

The reason for this finding is unclear, as this pattern of results does not seem to fit neatly into any existing literature on when interim testing may be harmful to future learning (e.g., Davis & Chan, 2015; Davis et al., 2017). One crucial distinction between the current study and past work is the provision of feedback; participants were given feedback following each interim test

in the procedure of Davis and Chan (2023) while the current study did not provide feedback to participants at any point. By eliciting several JOLs, including a JOL predicting performance on an immediate short-answer test, the test format used in the current study, learners may have made their predictions, taken the interim test, then remained confident in their performance on that test, experiencing less of a beneficial effect of testing than participants who made one or no JOLs. On the other hand, a simpler explanation is that tested participants who made four JOLs were simply less engaged with the material from the beginning, as evidenced by both the worse baseline performance on the first interim test relative to the other two JOL conditions and significantly lower average reading time on the criterial section relative to tested participants who made no JOLs.

When only a single JOL was elicited, tested participants still outperformed their untested counterparts on the criterial test; however, eliciting a single JOL caused participants who restudied the material to not significantly differ in performance from tested participants. These findings expand upon previous findings through the inclusion of a condition that restudied the material and a condition that made only a single JOL, with the data indicating that eliciting a single JOL may eliminate the FTE in untested participants who have a chance to restudy the material, but not those who do not. These results generally provide support for the metacognitive account for the FTE, as participants may be assumed to be engaging in the same introspection that is purported to occur in response to testing when making JOLs if they are given a restudy opportunity, prompting participants to change or maintain their encoding strategy over the course of study (Cho et al., 2017) or altering their expectations regarding a possible future test (Szpunar et al., 2007) after being given a chance to restudy the material. This is possibly due to the restudy opportunity serving as a chance for the learner to check how accurate they were when making

their metacognitive judgment by providing them with a chance to reexamine the material and determine how well they remembered it initially.

Contrary to the assumption of the metacognitive framework that testing benefits future learning at least in part due to interim tests curbing learners' overconfidence, criterial section JOLs did not differ between conditions in the current sample. This conclusion should be taken with a grain of salt, however, as determining whether overconfidence was curbed by testing would be better explored through examining metacognitive calibration (i.e., how accurate learners are in predicting their own performance; Pieschl, 2008), an analysis not conducted here due to scale differences between test performance and JOLs. Additionally, the current study provides reason to be wary of assuming the elicitation of JOLs to be a neutral event. The elimination of the forward benefits of testing when participants made several JOLs supports the notion that the elicitation of JOLs may induce reactivity, impacting memory performance, and thus may not be appropriate for use in testing paradigms as a neutral measure of metacognitive monitoring. As such, JOLs themselves are likely not appropriate to interpret as accurate indicators of prospective confidence when used in a retrieval practice context.

The current analyses did find some support for the metacognitive account in the reading time results as well; participants in the Test condition spent significantly more time reading the criterial section than participants in the other two interim task conditions, indicating a quantitative difference in encoding strategy for the final section, while there were no differences in reading time at baseline (i.e., Section 1). These results support the notion that interim testing may improve future test performance through providing the learner with metacognitive information that is then used to focus their effort and inform future study strategy use (i.e., either increasing the time they spend studying or keeping their study time/effort consistent over the

course of the study session while untested learners' study time may decrease; Davis & Chan, 2023; Yang et al., 2017).

Tested participants also generally reported much fewer extralist intrusions than untested participants in the current sample, providing support for the release-from-proactive interference account initially proposed by Szpunar et al. (2008) that was later incorporated into the 'context' explanatory framework. This framework posits that the act of retrieving material on an interim test induces a change in the internal context encoded with the material, allowing learners to constrain their search on future tests and experience less proactive interference (Kliegl et al., 2015; Szpunar et al., 2008; Wahlheim, 2015), enhancing memory outcomes. It is worth noting at this point that the mechanisms proposed by different explanations for the FTE are not necessarily mutually exclusive; each proposed mechanism may be more or less applicable in varying designs and contexts, while ultimately working in tandem to produce the benefits associated with interim testing. The data also support the notion that making a single JOL may prove beneficial in reducing proactive interference experienced by learners during study; however, this result is not theoretically expected and should be treated with a healthy degree of skepticism.

Contrary to expectations, there were differences in test performance at baseline, with participants in the Test condition who made no JOLs performing significantly better on the first interim test relative to Test condition participants who made JOLs. While the reason for this is unclear, one possible explanation stems from the fact that participants who made no JOLs were not given an additional filler task to be equated on time with those who made JOLs. It is possible, then, that participants who made no JOLs simply had the material decay less due to experiencing a shorter retention interval than participants who made JOLs, leading to better performance due to a lower delay between study and recall. Alternatively, it is possible that

participants in the JOL conditions experienced some degree of JOL reactivity prior to it being apparent on the criterial test, benefiting future learning enough to make differences between conditions null on the intermediary sections (e.g., Lee & Ha, 2019). It is worth stating here, however, that making comparisons between the first and later sections may be inappropriate given that the JOL conditions have diverged in the procedure by Section 2 and are no longer equivalent, possibly being differentially affected by the number of JOLs made on past sections.

The current study offers a practically viable look into the benefits of interpolated testing during study and JOL reactivity. In particular, the use of textbook passages and relatively simple benchmark questions that mirror those that one would expect to find in an introductory reading makes the results of this study relevant to academic contexts. Further, the current study is somewhat unique in its manipulation of JOLs and their inclusion in a typical FTE paradigm. Few studies at this point have explored the potential for the elicitation of JOLs to influence future learning outcomes (but see Lee & Ha, 2019), and fewer have done so using a typical testing paradigm with materials that one would expect to encounter in education like the ones used here and by Davis and Chan (2023). Additionally, in the case of Lee and Ha (2019), the reactivity effect observed was only explored in cases in which only JOLs were elicited, rather than in a retrieval practice context. Participants in the current study completed an interim task (filler, restudying, or taking a test) after making their JOL, providing further insight into the potential outcomes of combining JOL elicitation with other ways of interacting with the material.

While Davis and Chan (2023) also explored the possible additive effects of JOLs and interim testing, this study expanded upon their findings and design in multiple ways. First, the current study employed a factorial design to manipulate whether participants made JOLs within the same experiment, allowing for the direct examination of how JOLs may interact with interim

testing in their influence on criterial test performance. Second, the inclusion of a condition in which only a single JOL was elicited allowed for the investigation of whether one simple JOL would have the same reactivity effect that was observed when four JOLs eliciting more specific performance predictions were used by Davis and Chan (2023). The current study found that eliciting a single JOL only eliminated the benefits of testing when comparing tested participants with those that restudied the material. Finally, the current design included a Restudy condition. While most studies that investigate the FTE eschew the use of conditions which ask participants to restudy the material given their generally null results (e.g., Pastötter et al., 2011; Szpunar et al., 2008), the current study included such a condition to determine whether eliciting JOLs and thus alerting participants to how confident they feel in their comprehension of the material may cause them to use the restudy opportunity to check how well they actually remembered the passage, possibly eliciting a facilitative effect similar to that of interim testing.

While the current study benefited from its unique design and extension of prior work, there were several limitations that should be considered. First, the criterial test (and each interim test) was constrained in its variability due to the low number of questions employed on each test; each interim test, including the criterial test, only consisted of four cued-recall questions. As such, the distributions were relatively non-normal, having both ceiling and floor effects for each interim test, though this was less of a problem for the cumulative test data. Though ANOVAs are relatively robust to departures from normality (Blanca et al., 2017), future work may seek to remedy this limitation through using different materials.

As such, a second limitation of the current study concerns the materials used. While the current study made use of edited sections from a psychology textbook (Spielman et al., 2014), the text passage sections used were very brief – approximately 100 words in length each, much

shorter than what a student would typically be expected to read before encountering an interim test, diminishing the results' practical viability somewhat. In addition, the questions used here, partially adapted from Daley and Rawson (2019), had not been previously established and evaluated. Questions varied greatly in difficulty, with one interim test question in particular only being answered correctly by participants 22.1% of the time (compared to an average accuracy of 72.3% across interim tests). While this specific question being difficult is less problematic given that it did not appear on the criterial test, and benefits from retrieval practice have been found to occur regardless of how accurately the participants retrieve the material presented during interpolated tests (e.g., Richland et al., 2009), it is still worth mentioning, as it speaks to the variability in question and section difficulty. This variability makes it inappropriate for any comparisons to be made between sections.

A final limitation is the sample used in the present study. While the use of an online sample recruited via Prolific was convenient and allowed for a quick data collection process, the demographic diversity of the sample makes it difficult to generalize these results to the populations for whom they may be most important: students. Additionally, conducting online research runs into several problems intrinsic to online samples, such as sampling biases. Participants may be more likely to choose to participate in surveys that seem interesting to them or relevant to a topic they have experience with (Andrade, 2020). This issue may have been particularly prevalent in the current study, given that the topic of the study was memory and the text passage utilized in the study pertained to memory. Indeed, approximately 53% of the sample used in the current study indicated that they had previously taken a psychology course, where the topics covered in the text passage assuredly would have been covered given that they were adapted from an introductory psychology textbook (Spielman et al., 2014). It is worth reiterating,

however, that how well-versed participants were in the material prior to the study did not differ by condition. Additionally, use of an online sample disallows robust checks against cheating and makes it difficult for the experimenter(s) to be sure that the participant is providing the study with their full attention. This latter point is especially relevant given the use of reading time as a quantitative measure of encoding strategy, which may be less valid when using an online sample that is prone to getting distracted and/or leaving the page idle for extended periods. Future work in this area may seek to use collect data in a lab with students to overcome these limitations and extend these findings, gaining insight into how these effects may manifest in in-person settings with students specifically. Additionally, future research may look to use materials that participants are unlikely to have any experience with, such as text passages on niche topics (Davis & Chan, 2023), foreign word pairs (Cho et al., 2017), or novel face-name pairs (Weinstein et al., 2011).

Conclusions and Implications

In the present study, participants showed greater criterial test performance across JOL conditions if they completed interim tests after studying each section of a text passage, demonstrating an FTE. The metacognitive approach to the FTE posits that testing during a study session provides the learner with meaningful metacognitive information regarding how well the material is being learned and changing test expectations, with this information being used to guide future study decisions (Chan et al., 2018). In support of this approach, the analyses reported here showed that tested participants spent longer reading each text passage section on average and that eliciting JOLs, which may similarly induce introspection and provide the learner with useful metacognitive information, eliminated the FTE when participants were given a chance to restudy the judged material.

The current study provides further insight into the benefits of interim testing, as well as the theoretical mechanisms that contribute to them. This study is the first to manipulate the elicitation of JOLs alongside interpolated testing in a factorial design using educational materials. The results of the primary analysis were inconsistent with our hypothesis; however, exploratory analyses replicated past findings regarding the ability of metacognitive judgments to induce reactivity and eliminate the FTE (Davis & Chan, 2023). Critically, however, this reactivity was seemingly negative and experienced primarily by tested participants, who showed numerically lower criterial test performance when asked to make four JOLs after each of the first three sections. These results run contrary to both the metacognitive framework and the findings of Davis and Chan (2023), which posit that making multiple JOLs would benefit untested participants while leaving tested participants unaffected.

There are important theoretical and practical implications to be considered. First and foremost, the current study provides further support for the use of brief interim tests between textbook passages to enhance learning. Specifically, having learners complete a cued recall test after reading passages from a textbook should not only enhance learning of the tested passages, but also enhance learning of passages read after the test. Importantly, the use of such a varied adult sample may make the results generalizable to online workplace training; however, future work using video material that is more representative of the training videos used by employers is recommended to confirm this. An additional practical benefit of these findings is the lack of feedback required on interim tests to improve future learning outcomes; the act of retrieval was all that was necessary to produce an FTE in these data, though it is worth noting that Davis and Chan (2023) provided feedback during testing and found the same pattern of results. As such,

educators may feel freer to use cued recall questions during study without feeling dissuaded by a need to grade the responses.

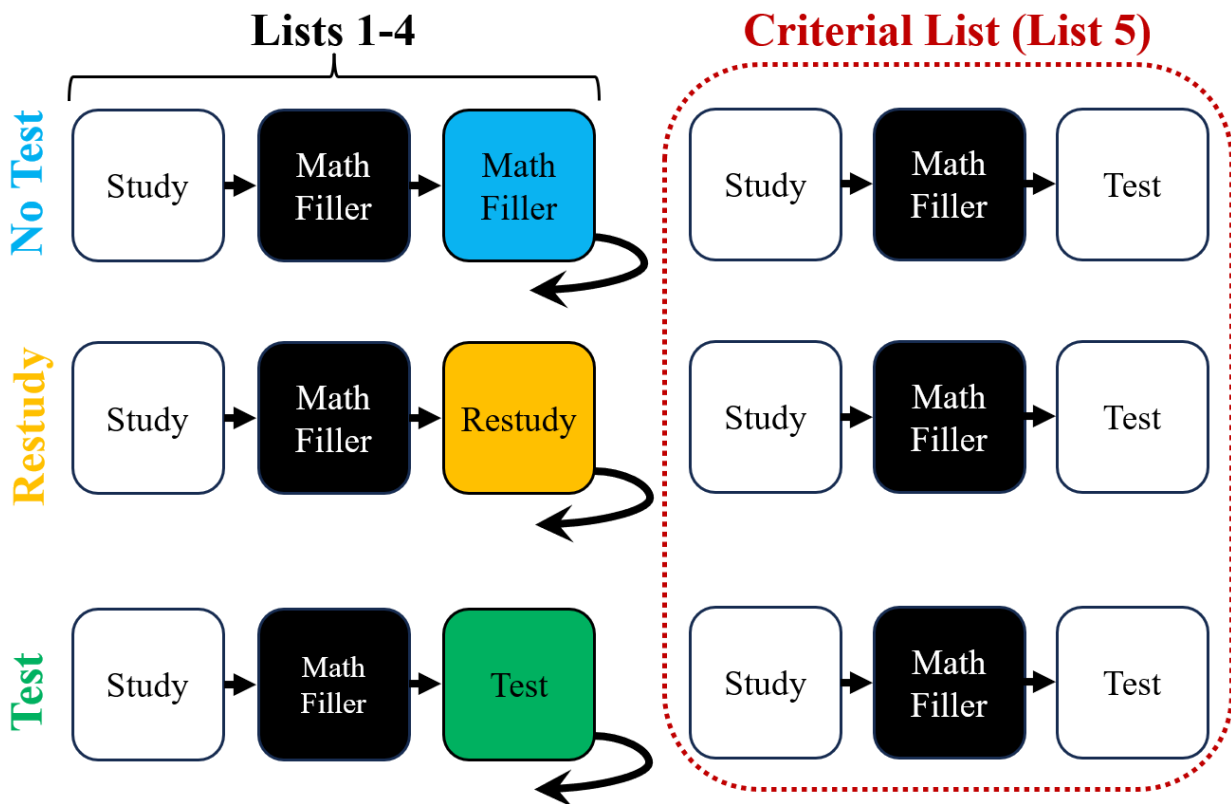
In conclusion, retrieval practice interleaved throughout a study session when reading an introductory textbook has been shown to be beneficial to future learning and delayed recall. Further support for the notions that interim testing causes learners to engage in more effective encoding strategies during future study and that testing serves to protect learners from the proactive interference that accumulates over the course of a study session has been demonstrated. Crucially, the current study has demonstrated that making multiple JOLs may be harmful to future learning when elicited in combination with interim testing during a study session, while eliciting JOLs and providing learners with a restudy opportunity directly after may be beneficial. While the elicitation of JOLs was found to eliminate the FTE under certain conditions in the current study, future work should seek to determine the contexts and procedures in which making metacognitive predictions of future performance may benefit, or impair, future study.

Table 1*Summary of Participant Demographics (N = 464)*

Variable	<i>n</i> (%)	<i>M</i> (<i>SD</i>)
Age (years)		41.86 (13.01)
Gender		
Male	232 (50.0%)	
Female	223 (48.1%)	
Nonbinary	6 (1.3%)	
Agender	1 (0.2%)	
Undisclosed	2 (0.4%)	
Race/Ethnicity		
Asian	30 (6.5%)	
Black or African American	55 (11.9%)	
Hispanic or Latino	11 (2.4%)	
Native Hawaiian or Pacific Islander	2 (0.4%)	
White	337 (72.6%)	
Multiracial	24 (5.1%)	
Undisclosed / Other	5 (1.1%)	
Education (Highest Attained)		
Some high school or less	4 (0.9%)	
High school diploma or GED	55 (11.0%)	
Some college, but no degree	85 (18.3%)	
Associate's or technical degree	49 (10.6%)	

Variable	<i>n</i> (%)	<i>M</i> (<i>SD</i>)
Bachelor's degree	185 (39.9%)	
Graduate or professional degree	90 (19.4%)	
Part Delay (hours)		23.59 (4.35)

Note. Part Delay refers to the time between participants finishing Parts 1 and 2 of the study.

Figure 1*Typical Forward Testing Effect Paradigm*

Note. Figure adapted from procedure for Szpunar et al. (2008) Experiment 3. Participants completed a short math filler task after each study session regardless of condition, with participants in the No Test condition completing the filler task again rather than taking the interim test or restudying.

Figure 2

Example of Response Excluded from Analyses for Suspected Cheating

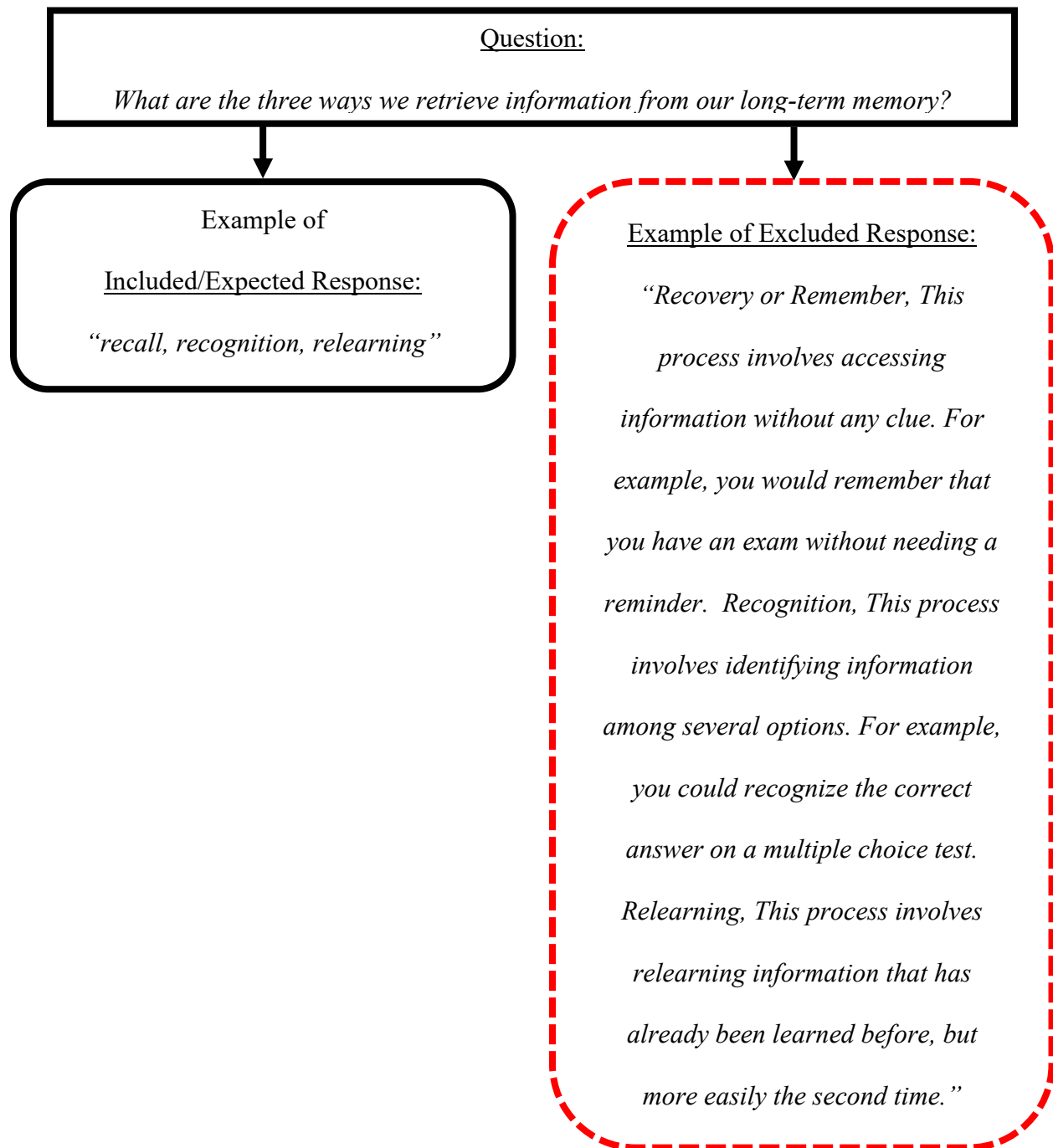
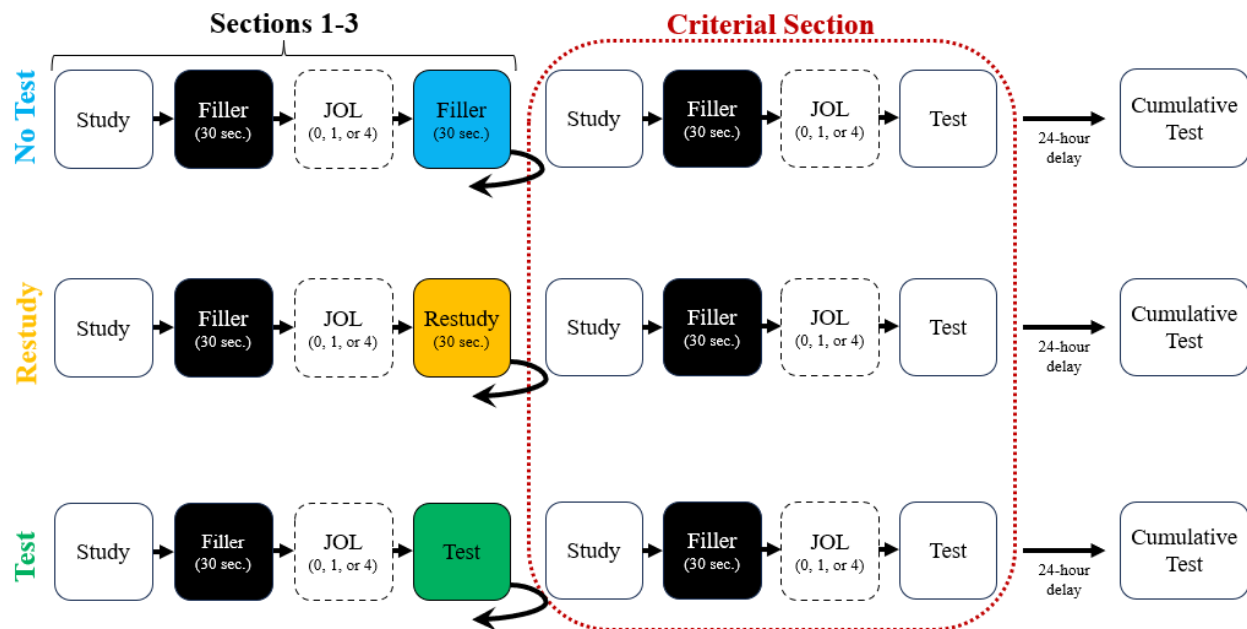
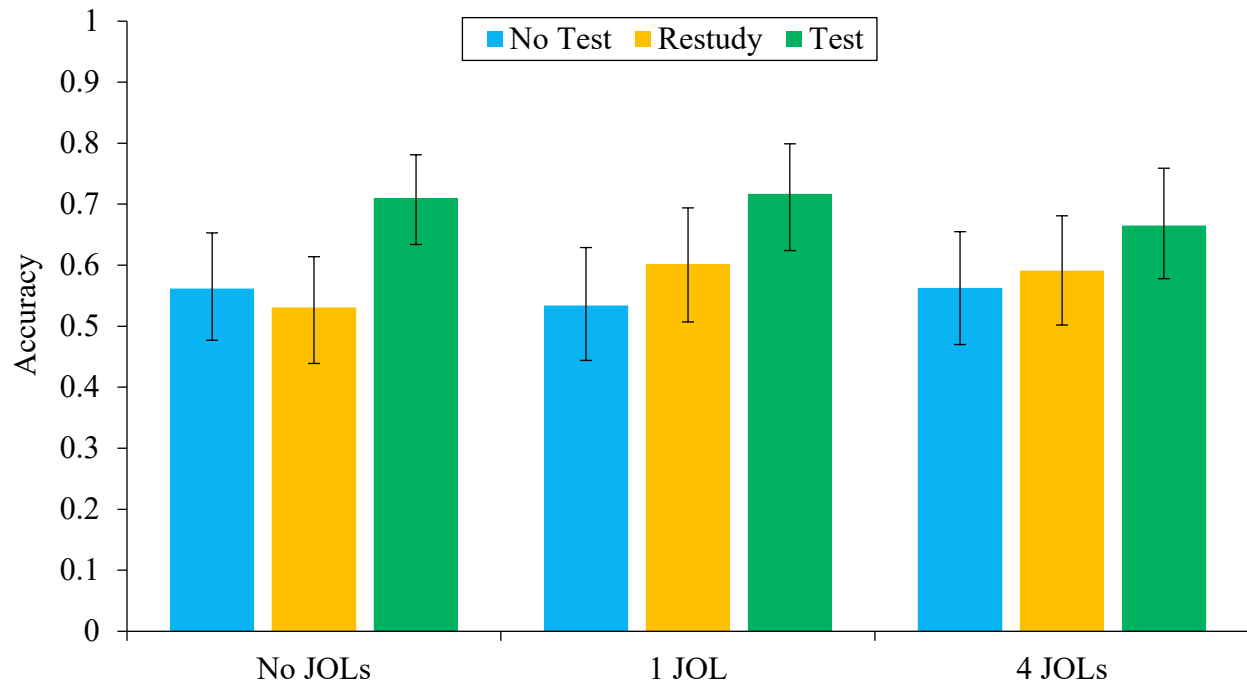


Figure 3*Procedure for the Present Study*

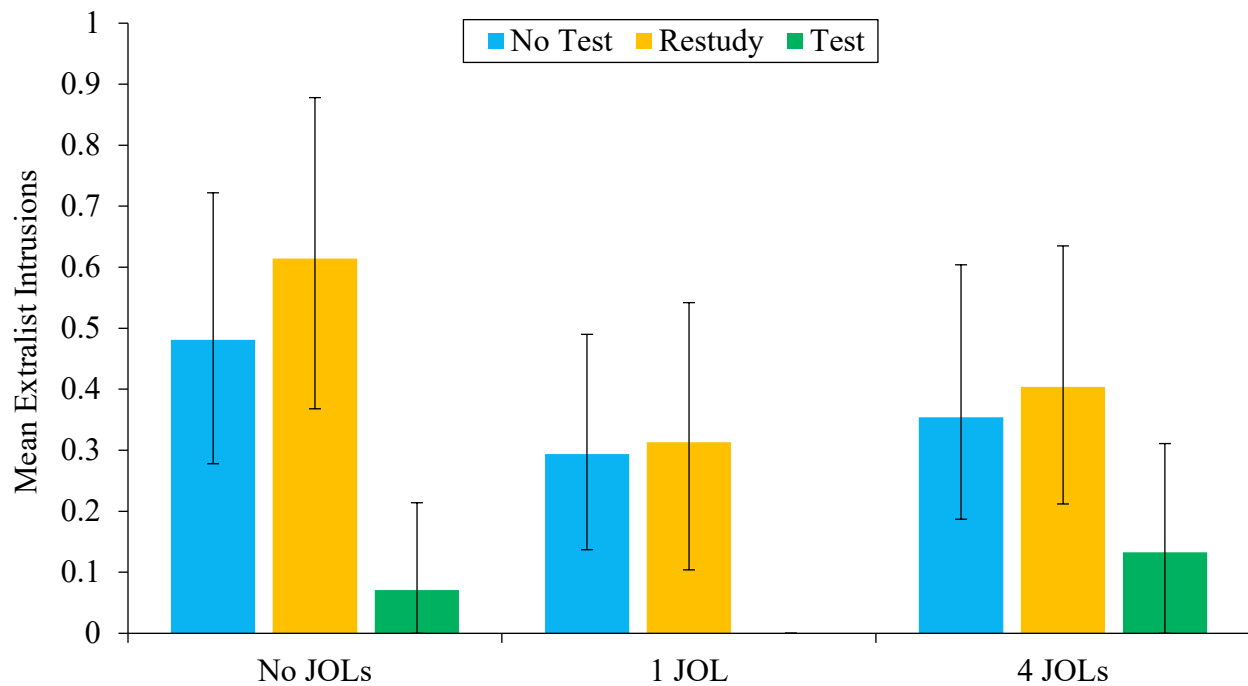
Note. The JOL task was omitted from the Critierial Section procedure for the first batch of participants due to experimenter error. Figure adapted from poster created using data from the first wave of data collection and presented at the annual meeting of the Psychonomic Society (2023).

Figure 4*Mean Criterial Test Performance by Condition*

Note. Error bars represent bootstrapped descriptive 95% confidence intervals about the mean.

Figure 5

Mean Criterial Section Extralist Intrusion Rates by Condition

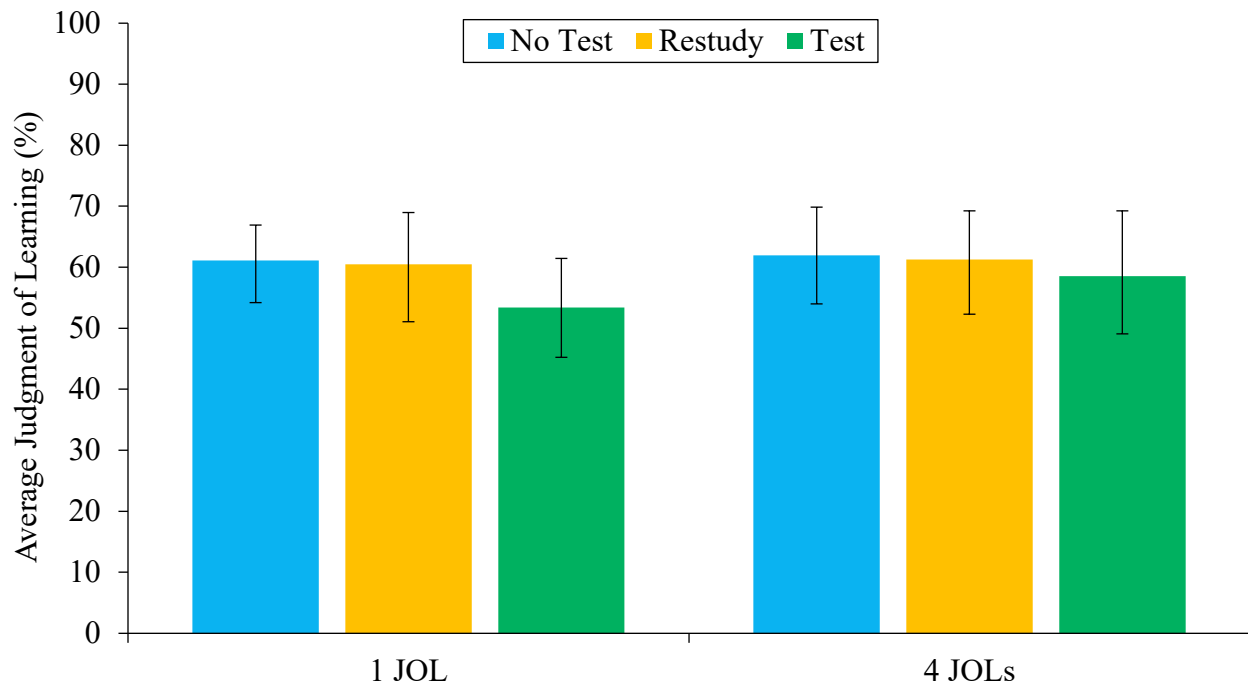


Note. Error bars represent bootstrapped descriptive 95% confidence intervals about the mean.

Participants in the Test + Single condition provided no responses that indicated extralist intrusions.

Figure 6

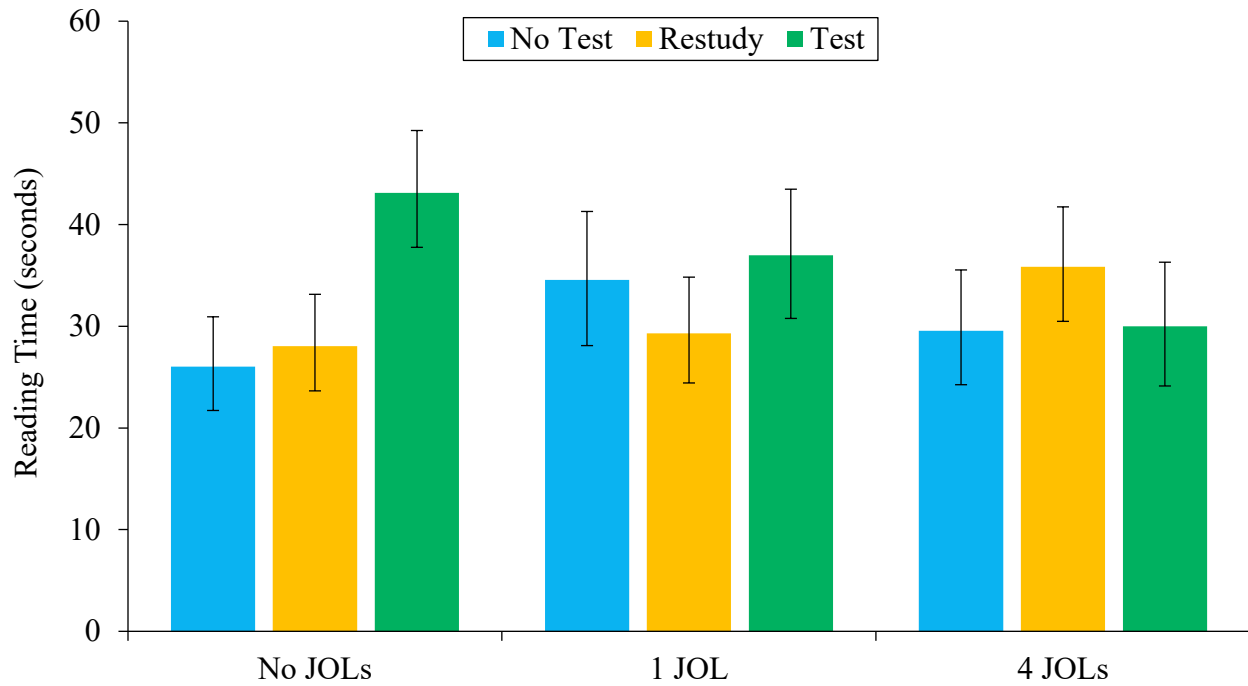
Mean Criterial Section Judgments of Learning by Condition



Note. Error bars represent bootstrapped descriptive 95% confidence intervals about the mean.

Figure 7

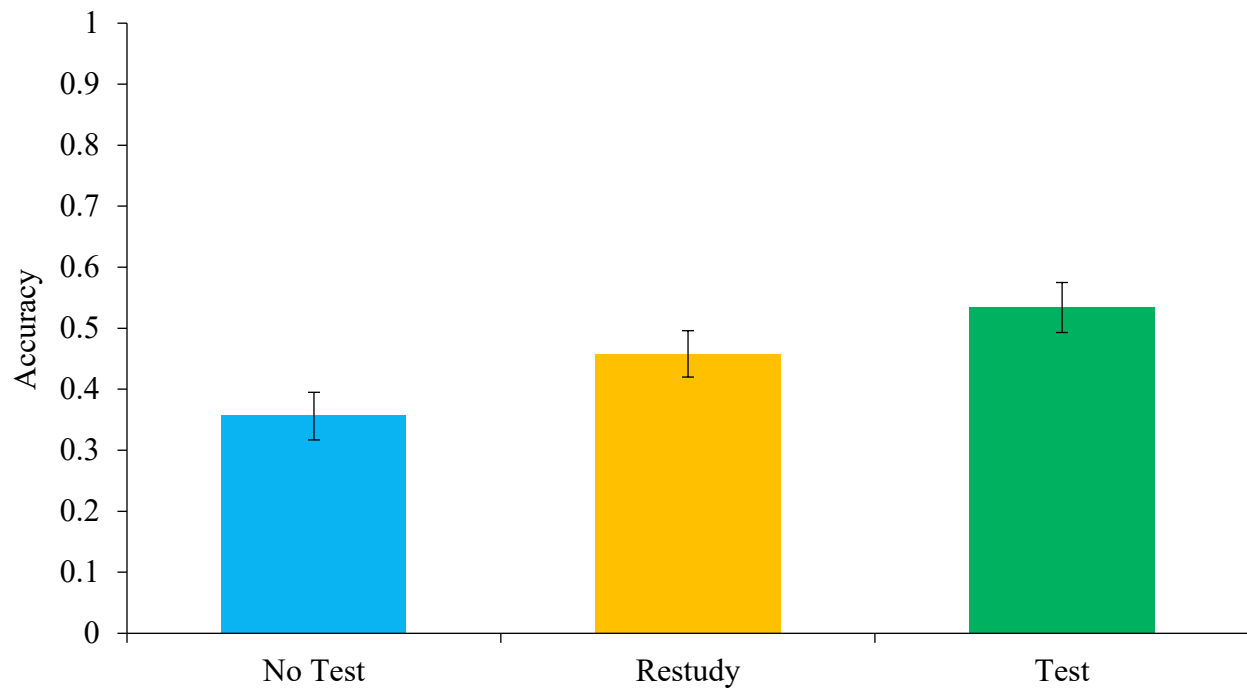
Mean Criterial Section Reading Time by Condition



Note. Error bars represent bootstrapped descriptive 95% confidence intervals about the mean.

Figure 8

Mean Cumulative Test Performance by Interim Task Condition



Note. Error bars represent bootstrapped descriptive 95% confidence intervals about the mean.

Means for each interim task condition are collapsed across the three JOL conditions.

Appendix A

Initial Pilot Results

To determine whether the materials used in the present study would be viable for investigation of the primary research questions, a preregistered pilot study was conducted using 60 participants recruited via the Prolific participant curation platform. The Qualtrics survey utilized in the pilot study was similar in structure to the final survey described in the paper; however, no JOLs were elicited from any participants in the pilot study. Participants were randomly assigned to one of three between-subjects conditions: Test, No Test, and Restudy. These conditions were identical to the interim task conditions used in the present study; Test condition participants were given a brief cued-recall test after reading each section, No Test participants completed a brief filler task, and Restudy participants were given 30 seconds to reread each section. The group of participants exposed to interim testing showed a numerically greater average proportion of items correctly recalled on the criterial test ($M = 0.63$, $SD = 0.34$) relative to both the group that had an opportunity to restudy the material between passages ($M = 0.45$, $SD = 0.36$) and the group that completed a filler task between passages ($M = 0.45$, $SD = 0.35$). These results were not statistically significant, $F(2, 57) = 1.73$, $p = .19$, $\eta_p^2 = .06$.

Appendix B

Text Passages and Questions

Passage 1 – Short-term memory (*109 words; Flesch-Kincaid Grade Level: 12.7*)

Short-term memory (STM) is a temporary storage system that processes incoming sensory memory; sometimes it is called working memory. Short-term memory storage lasts about 20 seconds. Think of short-term memory as the information you have displayed on your computer screen; a document, a spreadsheet, or a web page. Then, information in short-term memory goes to long-term memory (you save it to your hard drive), or it is discarded (you delete a document or close a web browser). The process of transferring information from short-term memory to long-term memory is called memory consolidation. Rehearsal is a form of memory consolidation and involves the conscious repetition of information to be remembered.

- 1) What type of memory acts as a temporary storage system? _____
- 2) How long does short-term memory last? _____
- 3) The "conscious repetition of information to be remembered" is known as what?

- 4) The process of moving information from short-term memory into long-term memory is known as what? _____

Passage 2 – Long-term memory (*102 words; Flesch-Kincaid Grade Level: 9.5*)

Long-term memory (LTM) is the continuous storage of information. Unlike short-term memory, the storage capacity of LTM has no limits. Long-term memory is divided into two types: implicit and explicit. Implicit memories are memories that are not part of our consciousness. They are memories formed from behaviors. Procedural memory is a type of implicit memory: it stores

information about how to do things. It is the memory for skilled actions, such as how to brush your teeth, how to drive a car, or how to swim the crawl (freestyle) stroke. These actions are practiced until they can be performed and remembered well.

- 1) What type of memory acts as our continuous storage of information? _____
- 2) What is the storage capacity of long-term memory? _____
- 3) What type of memory is not part of our consciousness? _____
- 4) What type of memory is responsible for storing information about how to do things like ride a bike? _____

Passage 3 – Explicit memory (*104 words; Flesch-Kincaid Grade Level: 13.1*)

Explicit memories are those we consciously try to remember and recall. Explicit memory has to do with the storage of facts and events we personally experienced. Explicit memory has two parts: semantic memory and episodic memory. Semantic means having to do with language and knowledge about language. Stored in our semantic memory is knowledge about words, concepts, language-based knowledge, and facts. Episodic memory is information about events we have personally experienced. Episodic memory is memory about happenings in particular places at

particular times, the what, where, and when of an event. It involves recollection of visual imagery as well as the feeling of familiarity.

- 1) What type of memories do we consciously try to remember and recall? _____
- 2) What is stored in our semantic memory? _____
- 3) What type of memory stores information about events we have personally experienced?

- 4) What feeling is associated with episodic memory? _____

Passage 4 – Retrieval (*101 words; Flesch-Kincaid Grade Level: 11.3*)

The act of getting information out of memory storage and back into conscious awareness is known as retrieval. There are three ways you can retrieve information out of your long-term memory storage system: recall, recognition, and relearning. Recall is what we most often think about when we talk about memory retrieval: it means you can access information without cues. Recognition happens when you identify information that you have previously learned after

encountering it again, such as on a multiple-choice test. It involves a process of comparison. The third form of retrieval is relearning. It involves learning information that you previously learned.

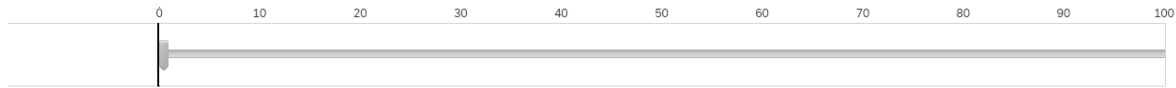
- 1) The act of getting information out of our memory and back into conscious awareness is known as what? _____
- 2) What are the three ways we retrieve information from our long-term memory? _____
- 3) What type of memory retrieval involves accessing information freely without the need for cues? _____
- 4) What form of retrieval involves identifying information that you have previously learned after encountering it again? _____

Appendix C

Judgments of Learning (JOLs)

Participants interacted with JOLs using a sliding scale from 0-100%; see example below.

From 0% correct to 100% correct, how well do you believe you would do on a test concerning the passage you just read? Please answer using the sliding scale below.



Single JOL Condition:

- 1) From 0% correct to 100% correct, how well do you believe you would do on a test concerning the passage you just read? Please answer using the sliding scale below.

Multiple JOLs Condition:

- 1) From 0% correct to 100% correct, how well do you believe you would do on a **multiple choice** test concerning the passage you just read if you were given one **right now**? Please answer using the sliding scale below.
- 2) From 0% correct to 100% correct, how well do you believe you would do on a **short answer** test concerning the passage you just read if you were given one **right now**? Please answer using the sliding scale below.
- 3) From 0% correct to 100% correct, how well do you believe you would do on a **multiple choice** test concerning the passage you just read if you were given one **25 minutes from now**? Please answer using the sliding scale below.
- 4) From 0% correct to 100% correct, how well do you believe you would do on a **short answer** test concerning the passage you just read if you were given one **25 minutes from now**? Please answer using the sliding scale below.

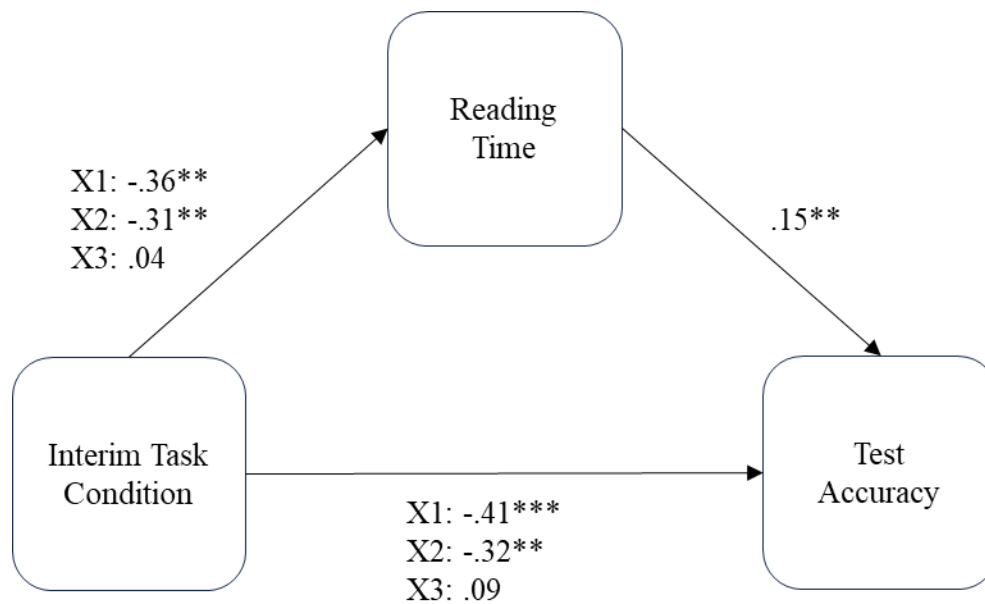
Appendix D

Effects of Interim Task Condition on Criterial Test Performance through Reading Time

As stated, it is possible that differences in reading time by interim task condition contribute indirectly to the observed differences in criterial test performance. To examine this possibility, mediation analyses examining the effect of interim task condition on performance through reading time were conducted using Model 4 of Hayes' PROCESS v4.1 for IBM SPSS (Hayes, 2022). Note that this model calculates indirect effects through multiplying the a and b path coefficients of the model ($a*b$) and constructs 95% confidence intervals via bootstrapping with 5000 resamples. In this model, interim task condition was used as a predictor, criterial section reading time as a mediator, and criterial section accuracy as an outcome variable. Interim task condition was dummy coded with the Test condition as the reference category. Additional analyses with the No Test condition as the reference category were conducted to determine whether there were any effects when comparing the No Test and Restudy conditions. Regarding the path coefficients, condition significantly predicted reading time ($F(2, 425) = 5.36, p = .005, R^2 = .02$), such that Test group membership was associated with higher reading times than membership in the No Test ($b = -7.22, \beta = -0.36, p = .003$) and Restudy ($b = -6.33, \beta = -0.31, p = .008$) conditions. The model also revealed significant differences in criterial test accuracy by condition and reading time ($F(3, 424) = 9.19, p < .001, R^2 = .06$). Differences in accuracy were found between Test and No Test participants ($b = -0.14, \beta = -0.41, p = .001$), as well between Test and Restudy participants ($b = -0.11, \beta = -0.32, p = .006$); participants in the Test condition could be expected to show higher test accuracy relative to participants in the No Test and Restudy conditions. Additionally, reading time was significantly positively associated with test accuracy ($b < .01, \beta = 0.15, p = .002$). As shown in the association between condition and

accuracy, there were direct effects of condition on test performance both when comparing Test to No Test (relative direct effect = -0.14, partially standardized relative direct effect = -0.41, 95% CI [-0.21, -0.06]) and Test to Restudy (relative direct effect = -0.11, partially standardized relative direct effect = -0.32, 95% CI [-0.18, -0.03]), $F(2, 424) = 6.75, p = .001, R^2\text{-change} = .03$. Participants in the Test condition can be expected to perform significantly better on the criterial test as a direct result of their condition. However, the mediation analyses also revealed significant indirect effects of condition on accuracy through reading time when comparing Test to No Test (relative indirect effect = -0.02, 95% CI [-0.04, -0.00]; partially standardized relative indirect effect = -0.05, 95% CI [-0.11, -0.01]) and when comparing Test to Restudy (relative indirect effect = -0.02, 95% CI [-0.03, -0.00]; partially standardized relative indirect effect = -0.05, 95% CI [-0.09, -0.01]), indicating that the relationship between interim testing and criterial test performance was partially mediated by reading time in the current sample. The mediation path model can be seen below. Another mediation analysis comparing No Test and Restudy found no significant differences between the two conditions; changes in condition were not associated with reading time ($b = 0.88, \beta = 0.04, p = .71$) nor test performance ($b = 0.03, \beta = 0.09, p = .43$); there were no direct effects of condition on criterial test accuracy with comparing No Test and Restudy (relative direct effect = 0.03, partially standardized relative direct effect = 0.09, 95% CI [-0.05, 0.10]), nor were there indirect effects on accuracy through reading time (relative indirect effect < 0.01, 95% CI [-0.01, 0.02]; partially standardized relative indirect effect = 0.01, 95% CI [-0.03, 0.05]). These results support the notion that testing informs future strategy use during study, with tested participants putting more time into encoding than untested participants. This interpretation should be accepted cautiously, however, considering the

problems associated with reading time as a metric in online studies where participant attention is subject to external distraction or disinterest.



Note. * $p < .05$, ** $p < .01$, *** $p < .001$. All coefficients are standardized (β). X1 represents the coefficient associated with the Test-No Test comparison, X2 represents the coefficient associated with the Test-Restudy comparison, X3 represents the coefficient associated with the No Test-Restudy comparison; all comparisons dummy coded with the first condition of the pair as the reference category.

References

- Abbott, E. E. (1909). On the analysis of the factor of recall in the learning process. *The Psychological Review: Monograph Supplements*, 11(1), 159–177.
<https://doi.org/10.1037/h0093018>
- Ahn, D., & Chan, J. (2023). Does Testing Potentiate New Learning Because it Enables Learners to Use Better Strategies? *Journal of Experimental Psychology Learning Memory and Cognition*.
- American Psychological Association. (2017). Ethical principles of psychologists and code of conduct (2002, amended effective June 1, 2010, and January 1, 2017).
<https://www.apa.org/ethics/code/>
- Andrade, C. (2020). The limitations of online surveys. *Indian Journal of Psychological Medicine*, 42(6), 575–576. <https://doi.org/10.1177/0253717620957496>
- Ariel, R., Karpicke, J. D., Witherby, A. E., & Tauber, S. K. (2021). Do Judgments of Learning Directly Enhance Learning of Educational Materials? *Educational Psychology Review*, 33(2), 693–712. <https://doi.org/10.1007/s10648-020-09556-8>
- Aslan, A., & Bäuml, K.-H. T. (2016). Testing enhances subsequent learning in older but not in younger elementary school children. *Developmental Science*, 19(6), 992–998.
<https://doi.org/10.1111/desc.12340>
- Bishara, A. J., & Jacoby, L. L. (2008). Aging, spaced retrieval, and inflexible memory performance. *Psychonomic Bulletin & Review*, 15(1), 52–57.
<https://doi.org/10.3758/PBR.15.1.52>

Blanca, M. J., Alarcón, R., Arnau, J., Bono, R., & Bendayan, R. (2017). Non-normal data: Is ANOVA still a valid option? *Psicothema*, 29(4), 552–557.

<https://doi.org/10.7334/psicothema2016.383>

Camp, C.J., Bird, M.J., & Cherry, K.E. (2000). *Retrieval strategies as a rehabilitation aid for cognitive loss in pathological aging*. In R.D. Hill, L. Backman, & A.S. Neely (Eds.), *Cognitive rehabilitation in old age* (pp. 224–248). New York: Oxford University Press.

Carpenter, S. K. (2011). Semantic information activated during retrieval contributes to later retention: Support for the mediator effectiveness hypothesis of the testing effect. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 37(6), 1547–1552.

<https://doi.org/10.1037/a0024140>

Chan, J. C. K., Manley, K. D., & Ahn, D. (2020). Does retrieval potentiate new learning when retrieval stops but new learning continues? *Journal of Memory and Language*, 115, 104150. <https://doi.org/10.1016/j.jml.2020.104150>

Chan, J. C. K., Meissner, C. A., & Davis, S. D. (2018). Retrieval potentiates new learning: A theoretical and meta-analytic review. *Psychological Bulletin*, 144, 1111–1146.

<https://doi.org/10.1037/bul0000166>

Cho, K. W., Neely, J. H., Crocco, S., & Vitrano, D. (2017). Testing enhances both encoding and retrieval for both tested and untested items. *Quarterly Journal of Experimental Psychology*, 70(7), 1211–1235. <https://doi.org/10.1080/17470218.2016.1175485>

Daley, N., & Rawson, K. A. (2019). Elaborations in Expository Text Impose a Substantial Time Cost but Do Not Enhance Learning. *Educational Psychology Review*, 31(1), 197–222.

<https://doi.org/10.1007/s10648-018-9451-9>

- Dang, X., Yang, C., Che, M., Chen, Y., & Yu, X. (2021). Developmental trajectory of the forward testing effect: The role of reset-of-encoding. *European Journal of Developmental Psychology*, 1–16. <https://doi.org/10.1080/17405629.2021.1986386>
- Davis, S., Chan, J., & Wilford, M. (2017). The Dark Side of Interpolated Testing: Frequent Switching Between Retrieval and Encoding Impairs New Learning. *Journal of Applied Research in Memory and Cognition*. <https://doi.org/10.1016/j.jarmac.2017.07.002>
- Davis, S. D., & Chan, J. C. K. (2023). Effortful Tests and Repeated Metacognitive Reflection Enhance Future Learning. *Educational Psychology Review*, 35(3). <https://doi.org/10.1007/s10648-023-09803-8>
- Davis, S. D., & Chan, J. C. K. (2015). Studying on borrowed time: How does testing impair new learning? *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 41, 1741–1754. <https://doi.org/10.1037/xlm0000126>
- Double, K. S., & Birney, D. P. (2017). Are you sure about that? Eliciting confidence ratings may influence performance on Raven’s progressive matrices. *Thinking & Reasoning*, 23(2), 190–206. <https://doi.org/10.1080/13546783.2017.1289121>
- Double, K. S., Birney, D. P., & Walker, S. A. (2018). A meta-analysis and systematic review of reactivity to judgements of learning. *Memory*, 26(6), 741–750. <https://doi.org/10.1080/09658211.2017.1404111>
- Dunlosky, J., & Rawson, K. A. (2012). Overconfidence produces underachievement: Inaccurate self evaluations undermine students’ learning and retention. *Learning and Instruction*, 22(4), 271–280. <https://doi.org/10.1016/j.learninstruc.2011.08.003>
- Dunn, P.K. (2021). *Scientific Research and Methodology: An introduction to quantitative research in science and health*. <https://bookdown.org/pkaldunn/Book>

- Evans, L. H., Williams, A. N., & Wilding, E. L. (2015). Electrophysiological evidence for retrieval mode immediately after a task switch. *NeuroImage*, 108, 435–440.
<https://doi.org/10.1016/j.neuroimage.2014.12.068>
- Finn, B., & Roediger III, H. L. (2013). Interfering effects of retrieval in learning new information. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 39, 1665–1681. <https://doi.org/10.1037/a0032377>
- Ford, C. (2015, August 26). *Understanding Q-Q Plots*. University of Virginia Library Research Data Services + Sciences. <https://data.library.virginia.edu/understanding-q-q-plots/>
- French, D. P., & Sutton, S. (2010). Reactivity of measurement in health psychology: How much of a problem is it? What can be done about it? *British Journal of Health Psychology*, 15(3), 453–468. <https://doi.org/10.1348/135910710x492341>
- Glantz, S. A., Slinker, B. K., & Neilands, T. B. (2016). *Primer of Applied Regression & Analysis of Variance, Third Edition*. McGraw-Hill Education / Medical.
- Halamish, V., & Undorf, M. (2022). Why do judgments of learning modify memory? Evidence from identical pairs and relatedness judgments. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, No Pagination Specified-No Pagination Specified.
<https://doi.org/10.1037/xlm0001174>
- Hayes, A. F. (2022). *Introduction to mediation, moderation, and conditional process analysis: A Regression-Based Approach*. Guilford Publications.
- Healy, A. F., Jones, M., Lalchandani, L. A., & Tack, L. A. (2017). Timing of quizzes during learning: Effects on motivation and retention. *Journal of Experimental Psychology: Applied*, 23(2), 128–137. <https://doi.org/10.1037/xap0000123>

- Jacoby, L.L. (1978). On interpreting the effects of repetition: Solving a problem versus remembering a solution. *Journal of Verbal Learning and Verbal Behavior*, 17, 649–667. [https://doi.org/10.1016/S0022-5371\(78\)90393-6](https://doi.org/10.1016/S0022-5371(78)90393-6)
- JASP Team (2024). JASP (Version 0.18.3) [Computer software].
- Jing, H. G., Szpunar, K. K., & Schacter, D. L. (2016). Interpolated testing influences focused attention and improves integration of information during a video-recorded lecture. *Journal of Experimental Psychology: Applied*, 22, 305–318. <https://doi.org/10.1037/xap0000087>
- Karaca, M., Kurpad, N., Wilford, M. M., & Davis, S. L. (2020). Too much of a good thing: frequent retrieval can impair immediate new learning. *Memory*, 28(10), 1181–1190. <https://doi.org/10.1080/09658211.2020.1826526>
- Karpicke, J. D. (2009). Metacognitive control and strategy selection: Deciding to practice retrieval during learning. *Journal of Experimental Psychology: General*, 138(4), 469–486. <https://doi.org/10.1037/a0017341>
- Kliegl, O., Pastötter, B., & Bäuml, K.-H. T. (2015). The contribution of encoding and retrieval processes to proactive interference. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 41(6), 1778–1789. <https://doi.org/10.1037/xlm0000096>
- Kornell, N. (2014). Attempting to answer a meaningful question enhances subsequent learning even when feedback is delayed. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 40, 106–114. <https://doi.org/10.1037/a0033699>
- Kornell, N., Hays, M. J., & Bjork, R. A. (2009). Unsuccessful retrieval attempts enhance subsequent learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 35, 989–998. <https://doi.org/10.1037/a0015729>

- Kubik, V., Koslowski, K., Schubert, T., & Aslan, A. (2022). Metacognitive judgments can potentiate new learning: The role of covert retrieval. *Metacognition and Learning*, 17, 1057-1077. <https://doi.org/10.1007/s11409-022-09307-w>
- Lau, M. (2022). *Student Testing: An International Context*. Fraser Institute.
<https://www.fraserinstitute.org/studies/student-testing-an-international-context>
- Lee, H. S., & Ahn, D. (2018). Testing prepares students to learn better: The forward effect of testing in category learning. *Journal of Educational Psychology*, 110, 203–217.
<https://doi.org/10.1037/edu0000211>
- Lee, H. S., & Ha, H. (2019). Metacognitive judgments of prior material facilitate the learning of new material: The forward effect of metacognitive judgments in inductive learning. *Journal of Educational Psychology*, 111, 1189–1201.
<https://doi.org/10.1037/edu0000339>
- Lee, S., & Lee, D. K. (2018). What is the proper way to apply the multiple comparison test? *Korean Journal of Anesthesiology*, 71(5), 353–360.
<https://doi.org/10.4097/kja.d.18.00242>
- Leeming, F. C. (2002). The exam-a-day procedure improves performance in psychology classes. *Teaching of Psychology*, 29(3), 210–212.
https://doi.org/10.1207/S15328023TOP2903_06
- McDermott, K. B. (2021). Practicing Retrieval Facilitates Learning. *Annual Review of Psychology*, 72(1), 609–633. <https://doi.org/10.1146/annurev-psych-010419-051019>
- Metcalf, J., & Finn, B. (2008). Evidence that judgments of learning are causally related to study choice. *Psychonomic Bulletin & Review*, 15(1), 174–179.
<https://doi.org/10.3758/PBR.15.1.174>

- Myers, S. J., Rhodes, M. G., & Hausman, H. E. (2020). Judgments of learning (JOLs) selectively improve memory depending on the type of test. *Memory & Cognition*, 48(5), 745–758. <https://doi.org/10.3758/s13421-020-01025-5>
- Nelson, T. O., & Narens, L. (1990). Metamemory: A Theoretical Framework and New Findings. In *Psychology of Learning and Motivation* (Vol. 26, pp. 125–173). Elsevier. [https://doi.org/10.1016/S0079-7421\(08\)60053-5](https://doi.org/10.1016/S0079-7421(08)60053-5)
- Nunes, L. D., & Weinstein, Y. (2012). Testing improves true recall and protects against the build-up of proactive interference without increasing false recall. *Memory*, 20(2), 138–154. <https://doi.org/10.1080/09658211.2011.648198>
- Pastötter, B., Schicker, S., Niedernhuber, J., & Bäuml, K.-H. T. (2011). Retrieval during learning facilitates subsequent memory encoding. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 37, 287–297. <https://doi.org/10.1037/a0021801>
- Pastötter, B., Weber, J., & Bäuml, K.-H. T. (2013). Using testing to improve learning after severe traumatic brain injury. *Neuropsychology*, 27, 280–285. <https://doi.org/10.1037/a0031797>
- Pieschl, S. (2008). Metacognitive calibration—an extended conceptualization and potential applications. *Metacognition and Learning*, 4(1), 3–31. <https://doi.org/10.1007/s11409-008-9030-4>
- Rhodes, M. G. (2015). *Judgments of Learning* (J. Dunlosky & S. (Uma) K. Tauber, Eds.; Vol. 1). Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780199336746.013.4>
- Rivers, M. L., Janes, J. L., & Dunlosky, J. (2021). Investigating memory reactivity with a within-participant manipulation of judgments of learning: Support for the cue-strengthening

hypothesis. *Memory*, 29(10), 1342–1353.

<https://doi.org/10.1080/09658211.2021.1985143>

Roediger, H. L., & Butler, A. C. (2011). The critical role of retrieval practice in long-term retention. *Trends in Cognitive Sciences*, 15(1), 20–27.

<https://doi.org/10.1016/j.tics.2010.09.003>

Roediger, H. L., & Karpicke, J. D. (2006a). Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention. *Psychological Science*, 17(3), 249–255.

<https://doi.org/10.1111/j.1467-9280.2006.01693.x>

Roediger, H. L., & Karpicke, J. D. (2006b). The Power of Testing Memory: Basic Research and Implications for Educational Practice. *Perspectives on Psychological Science*, 1(3), 181–210. <https://doi.org/10.1111/j.1745-6916.2006.00012.x>

Rohrer, D., Taylor, K., & Sholar, B. (2010). Tests enhance the transfer of learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 36(1), 233–239.

<https://doi.org/10.1037/a0017678>

Rowland, C. A. (2014). The effect of testing versus restudy on retention: A meta-analytic review of the testing effect. *Psychological Bulletin*, 140(6), 1432–1463.

<https://doi.org/10.1037/a0037559>

Schacter, D. L., Rich, S. A., & Stamp, M. S. (1985). Remediation of memory disorders: Experimental evaluation of the Spaced-Retrieval technique. *Journal of Clinical and Experimental Neuropsychology*, 7(1), 79–96.

<https://doi.org/10.1080/01688638508401243>

- Smallwood, J., & Schooler, J. W. (2015). The Science of Mind Wandering: Empirically Navigating the Stream of Consciousness. *Annual Review of Psychology*, 66(1), 487–518. <https://doi.org/10.1146/annurev-psych-010814-015331>
- Soderstrom, N. C., Clark, C. T., Halamish, V., & Bjork, E. L. (2015). Judgments of learning as memory modifiers. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 41(2), 553–558. <https://doi.org/10.1037/a0038388>
- Spielman, R. M., Dumper, K., Jenkins, W., Lacombe, A., Lovett, M., & Perlmutter, M. (2014). *How Memory Functions. In Psychology*. OpenStax. <https://openstax.org/books/psychology/pages/8-1-how-memory-functions>
- Szpunar, K. K., Jing, H. G., & Schacter, D. L. (2014). Overcoming overconfidence in learning from video-recorded lectures: Implications of interpolated testing for online education. *Journal of Applied Research in Memory and Cognition*, 3, 161–164. <https://doi.org/10.1016/j.jarmac.2014.02.001>
- Szpunar, K. K., Khan, N. Y., & Schacter, D. L. (2013). Interpolated memory tests reduce mind wandering and improve learning of online lectures. *Proceedings of the National Academy of Sciences*, 110(16), 6313–6317. <https://doi.org/10.1073/pnas.1221764110>
- Szpunar, K. K., McDermott, K. B., & Roediger, H. L., III. (2007). Expectation of a final cumulative test enhances long-term retention. *Memory & Cognition*, 35, 1007–1013. <http://dx.doi.org/10.3758/BF03193473>
- Szpunar, K. K., McDermott, K. B., & Roediger III, H. L. (2008). Testing during study insulates against the buildup of proactive interference. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 34, 1392–1399. <https://doi.org/10.1037/a0013082>

- Thomas, R. C., & McDaniel, M. A. (2013). Testing and feedback effects on front-end control over later retrieval. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 39(2), 437–450. <https://doi.org/10.1037/a0028886>
- Tulving, E. (1983). *Elements of Episodic Memory*. Oxford University Press.
- Tulving, E., & Watkins, M. T. (1974). On negative transfer: Effects of testing one list on the recall of another. *Journal of Verbal Learning and Verbal Behavior*, 13(2), 181–193. [https://doi.org/10.1016/s0022-5371\(74\)80043-5](https://doi.org/10.1016/s0022-5371(74)80043-5)
- Wahlheim, C. N. (2015). Testing can counteract proactive interference by integrating competing information. *Memory & Cognition*, 43(1), 27–38. <https://doi.org/10.3758/s13421-014-0455-5>
- Wang, X., Liu, X., Chen, L., Feng, K., Ye, Q., & Zhu, H. (2023). The forward effect of delayed judgments of learning is influenced by difficulty in memory and category learning. *Journal of Intelligence*, 11(6), 101. <https://doi.org/10.3390/jintelligence11060101>
- Weinstein, Y., McDermott, K. B., & Szpunar, K. K. (2011). Testing protects against proactive interference in face–name learning. *Psychonomic Bulletin & Review*, 18(3), 518–523. <https://doi.org/10.3758/s13423-011-0085-x>
- Wissman, K. T., Rawson, K. A., & Pyc, M. A. (2011). The interim test effect: Testing prior material can facilitate the learning of new material. *Psychonomic Bulletin & Review*, 18(6), 1140–1147. <https://doi.org/10.3758/s13423-011-0140-7>
- Yang, C., Potts, R., & Shanks, D. R. (2017). The forward testing effect on self-regulated study time allocation and metamemory monitoring. *Journal of Experimental Psychology: Applied*, 23, 263–277. <https://doi.org/10.1037/xap0000122>

Yang, C., Potts, R., & Shanks, D. R. (2018). Enhancing learning and retrieval of new information: A review of the forward testing effect. *npj Science of Learning*, 3(1).

<https://doi.org/10.1038/s41539-018-0024-y>

Zhou, A., Yang, T., Cheng, C., Ma, X. & Zhao, J. (2015). Retrieval practice produces more learning in multiple-list tests with higher-order skills. *Acta Psychol. Sin.* 47, 928.

<https://doi.org/10.3724/SP.J.1041.2015.00928>